Surgical robotics today

Who are all these companies & How did they get here?

Dwight Meglan

12Nov26 Update · The commercial surgical robotics landscape

Companion to the Surgical AI Overview (Fall 2026)

All information discussed is derived from publicly available sources

Why me for this subject?

30+ years building surgical robots and simulators, and tracking everyone else's

- **Left Mayo tenure track 30+ years ago to stay an engineer**
  Building surgical simulators since 1994; surgical robots since 2000
- **Part of many investment & acquisition diligence efforts**
  Surgical robotics and adjacent: robotic systems, components and enablers; advanced visualization; machine learning
- **20+ years tracking commercial robots & simulators**
  I work at the system level, from idea to prototype

15 COMMERCIAL SURGICAL ROBOTS

Involvement across specialties:

- 4 Abdominal / thoracic
- 4 Endovascular
- 3 Endoluminal
- 2 Cardiac
- 1 Ophthalmic
- 1 Vascular access

Why am I doing this?

A hobby that turned out to be useful, everything is a remix

- **I started collecting this information as a hobby**
  Knowing how similarly focused robots could have diverse solutions was useful for my design work (everything is a remix)
- **As the collection grew, others might find it useful**
  This deck is an overview of the field touching on the diversity of solutions for similar needs, e.g. surgeon consoles
- **I do deep dives for my own startup work and consulting**

NOTE

This talk is largely mechanics focused, but I typically work at the system level

- I work at the idea-to-prototype phase, i.e. typically much earlier than Steve Bell :-)

This talk…

A scope contract: breadth over depth, commercial over academic

IS

- A quick run through the breadth of commercial efforts
- An update on what has changed in the past ~6 months
- Examples of how this information provides insight
- Robots that act upon the body in some manner
- A broad view of 'commercial', even patents alone

IS NOT

- By any means exhaustive
- Delving into the myriad academic efforts
- Covering rehabilitation or assistive robots

Surgical robotics today: update notes, July 2026

What changed since the January 2026 edition

- **774 robots now, 658 in January**
  Roster 774 robots / 677 companies. 58 robots first seen in 2026: 9 orthopedic, 7 abdominal/thoracic, 6 percutaneous, 6 endoluminal, 4 spine… 34 of the 58 are Chinese, mostly EPO patent-scan finds, not press
- **Full roster re-verification, 1–2 July**
  532 field corrections, 121 new milestones, duplicates merged; quietly-approved systems re-checked, On Market count: 86 in January → 244 now
- **Funding data added for every robot**
  Stage, total raised, lead investors, valuation and most-recent-milestone columns now populated throughout

ALSO NEW IN THIS EDITION

- Funding & investor data throughout
- 2026 flameouts & exits section
- Founding-year (vs first-evidence) view
- All charts regenerated from the master roster
- Live companion dashboard, see the end of the deck

Big picture on surgical robotics

Note: most stats reflect robots, not robot companies: the roster is one row per robot

Surgical robotics is a busy place

316 review articles in 2020–2022 per the refreshed counts; a sampling of 2022 titles below

HISTORY & OVERVIEWS

A short history of robotic surgery · Past, present and future of surgical robotics · An introductory review of robotically assisted surgical systems

IMAGING & GUIDANCE

Image-Guided Interventional Robotics: Lost in Translation? · Robot-Assisted Medical Imaging: A Review · MRI-Guided Robot-Assisted Surgery and Interventions

SPECIALTY-SPECIFIC

Robotics in neurosurgery · Intraocular Robotic Surgical Systems · Robot-assisted techniques in vascular and endovascular surgery · Robotic Systems in Radiotherapy

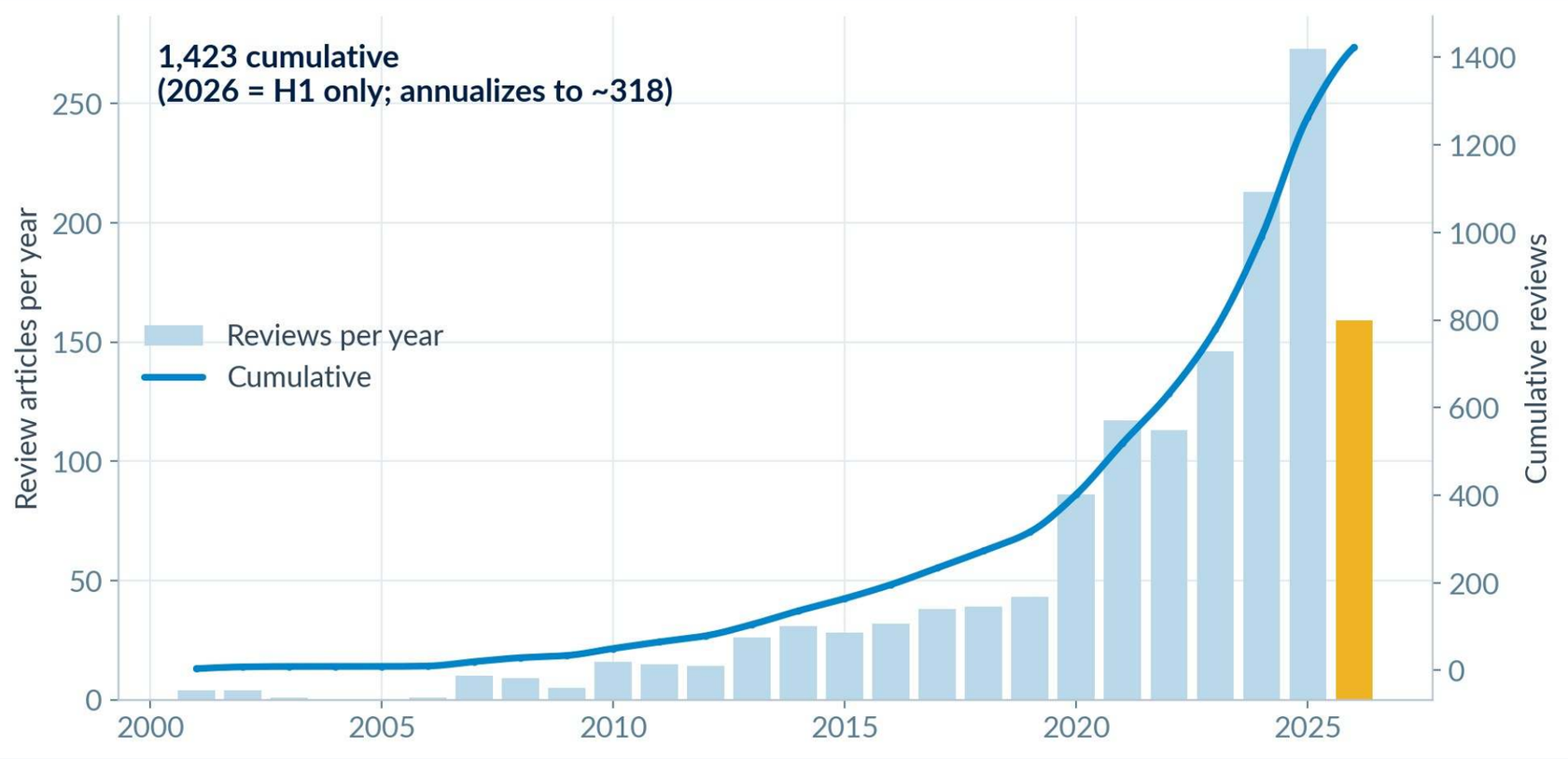
EMERGING TECH

Magnetically Actuated Medical Robots · Surgical Robotics in the Data Age · among others published in 2022

Source: Google Scholar, review-article searches on surgical robotics, 2020–2022.

Surgical robotics is a busy place

Review articles per year, Google Scholar, refreshed 7 Jul 2026; 2026 = first half, actual



1,423 REVIEWS

cumulative since 2001

273 in 2025; 159 in H1 2026, annualizing to roughly 318

Same Google Scholar query as prior editions; counts refreshed 7 Jul 2026

Where it came from

J Troccaz et al., Frontiers of Medical Robotics, Annu. Rev. Biomed. Eng. 21:193-218, 2019

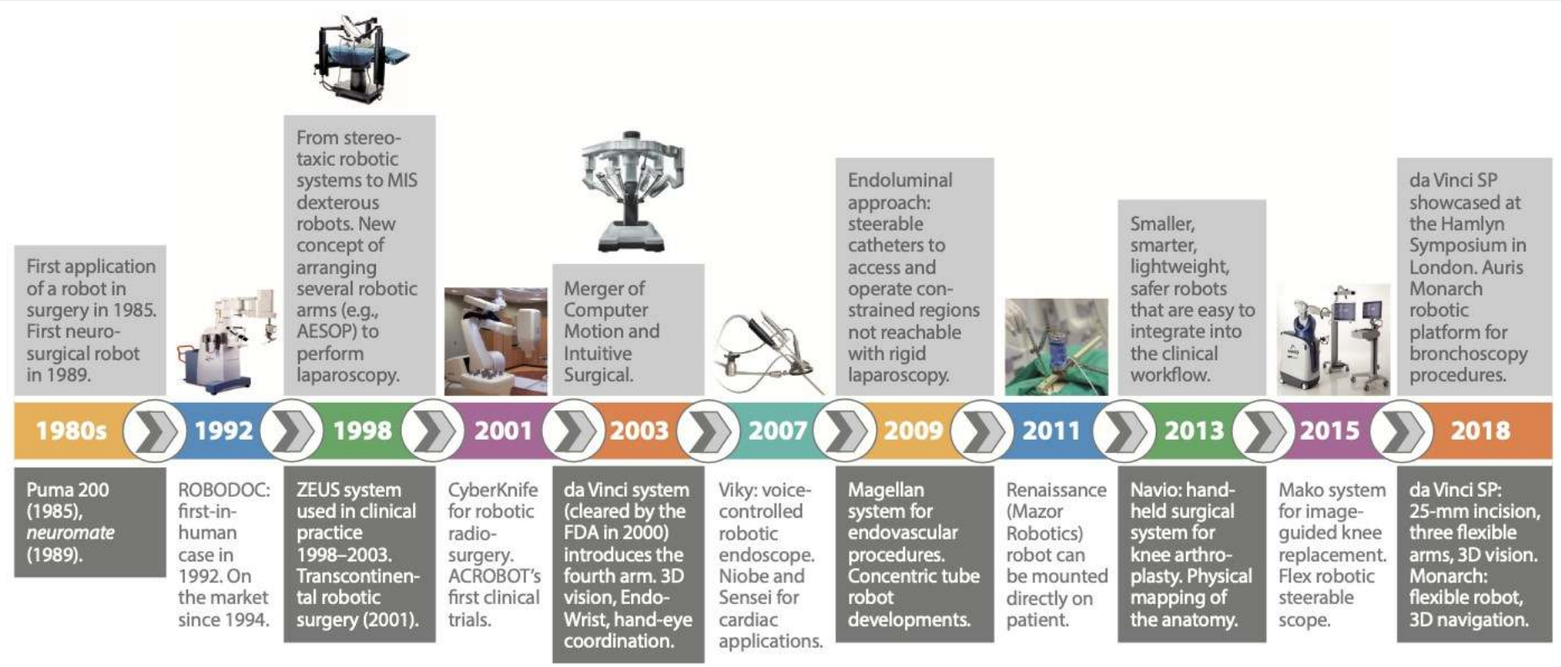


Figure 2

Timeline of medical robotics showing key milestones in medical robotics and computer-assisted intervention. Only commercial surgical platforms are depicted. Abbreviations: 3D, three-dimensional; FDA, Food and Drug Administration; MIS, minimally invasive surgery; SP, single port. Images are adapted with permission as follows: ROBODOC, Reference 130; ZEUS/AESOP, Reference 20; CyberKnife, Wikimedia Commons; da Vinci, Intuitive Surgical Inc., © 2018; Viky, Endocontrol, © 2018; Mazor Robotics Renaissance, Mazor Robotics; Mako, Reference 131.

J Troccaz et al., Frontiers of Medical Robotics: From Concept to Systems to Clinical Translation
Annual Review of Biomedical Engineering, Vol. 21:193-218, 2019

Where it came from

knowablemagazine.org, Handing the surgeon's scalpel to a robot (2022)

Timeline of progress in surgical robots



1970s NASA explores telesurgery.

1983 First robot in the surgical suite, used to move patient's limbs into position during orthopedic surgery.

1985 First instance of a surgical robot, used during brain biopsy to avoid errors from hand tremors.



1985 First laparoscopic procedure.

1990s Laparoscopic surgery becomes more common, many novel procedures carried out and many new systems invented.

1991 First telepresence system, means surgeons don't need to be directly next to patient.



1992 First robot with task autonomy gets FDA approval. Used in hip replacement, ROBODOC can carry out certain pre-programmed tasks autonomously once initiated by a human.

2000s Surgical robots for general use become more common.



2000 Introduction of da Vinci, the first robotic system for general and laparoscopic surgery in humans.

2001 Doctors in New York City use telesurgery to operate on a patient in France.



2001 First robot with conditional autonomy gets FDA approval. Used in radiation therapy, CyberKnife can perform procedure autonomously.

2001 – present Refinement of existing surgical robots continues to expand, robots become smaller, lighter, easier to use, and offer more assistance features for surgeons.

2009 Steerable catheters to access and operate constrained regions not reachable with rigid laparoscopy.

2022 STAR robot successfully sutures severed pig intestine.

The future Surgical robots gradually acquire increasing autonomy.

SOURCE: REPORTING BY J. GAINES

KNOWABLE MAGAZINE

https://knowablemagazine.org/article/technology/2022/handing-surgeon-scalpel-to-robot

Commercial milestones

T Ginoya et al., A Historical Review of Medical Robotic Platforms, J. Robotics 2021, doi.org/10.1155/2021/6640031

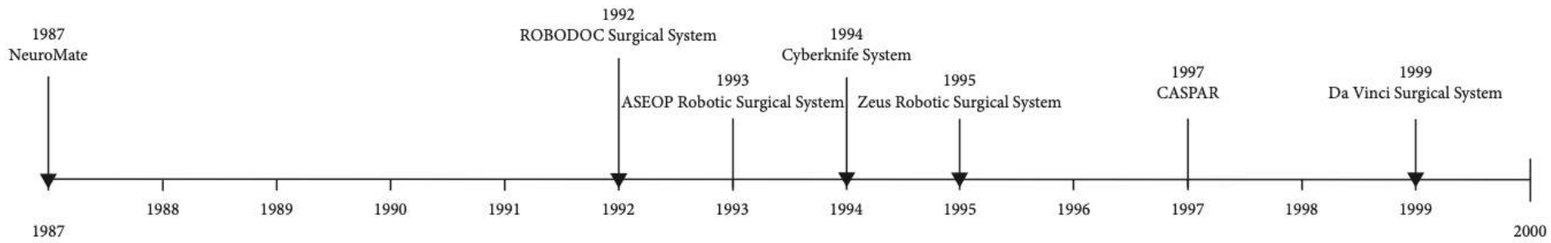


FIGURE 1: Decade I: timeline of robotic systems that founded and created the base for usage of robots in the medical field.

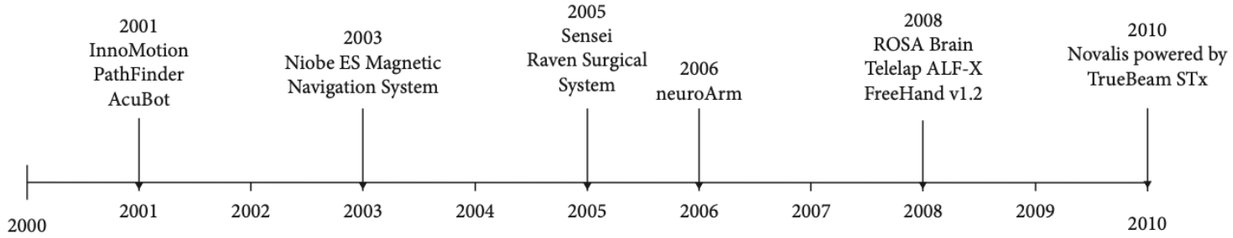


FIGURE 2: Decade II–timeline of robotic systems that signify the progress of the development curve of Robots in medical application.

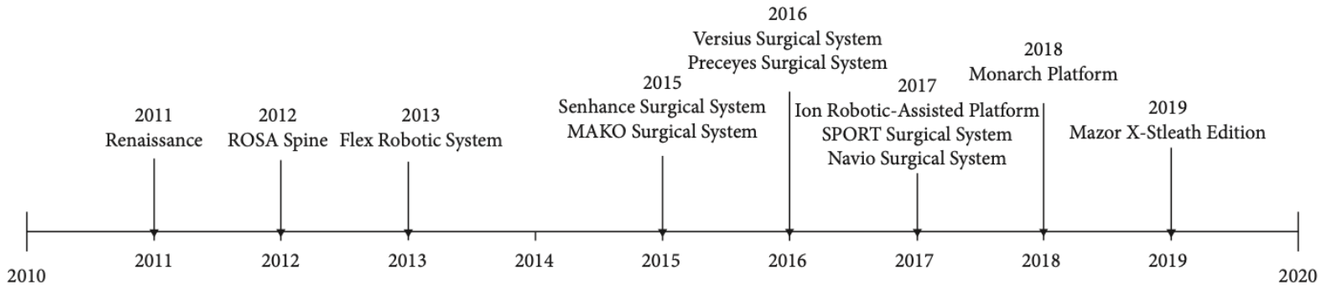


FIGURE 3: Decade III–timeline of robotic systems that show the recent advancement in the field of medical robots based on the advancement of technology and effort of three decades of research in this field.



Commercial surgical robotics trends

Five parts: the roster today · how fast it grew · where · specialty waves · the endovascular case.
Stats reflect robots, not companies.

Lots of bots, mostly on the way

Roster · the field at a glance; there are more: stealth and non-US companies evade detection

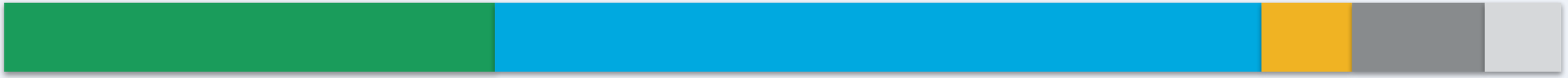
774
robots tracked

677
companies

244
on market

67
multi-robot companies

STATUS OF THE 774 ROBOTS

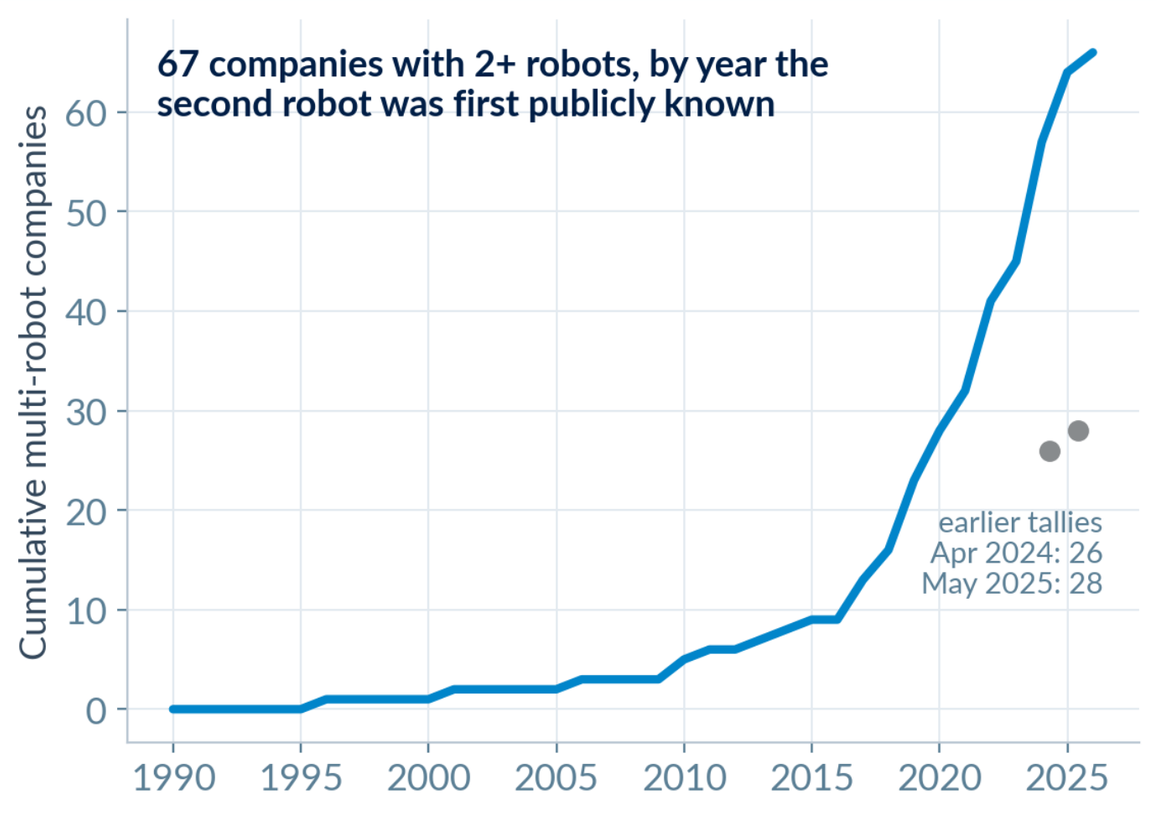


On Market (244) Under Dev (381) Acquired (45) Dead (66) Uncertain (38)

On Market re-verified against regulator databases & company disclosures. Rows are robots; companies counted once across multi-robot portfolios.

Multi-robot portfolios are mostly recent

Roster · sources: patent filings, FDA/NMPA submissions, press releases, papers; collected 8-11 Jul 2026



67 MULTI-ROBOT COMPANIES

10% of tracked companies field two or more robots

Dated by first public evidence for each second robot: patent filing, FDA/NMPA submission, press release, or paper

57% went multi-robot in 2021 or later; earlier edition tallies (26, 28) lagged detection

How they work

Roster · control philosophy across all 774 robots; teleoperation still dominates

488
teleoperation

176
preplanned

82
autonomous

28
handheld & other

CONTROL MODE OF THE 774 ROBOTS

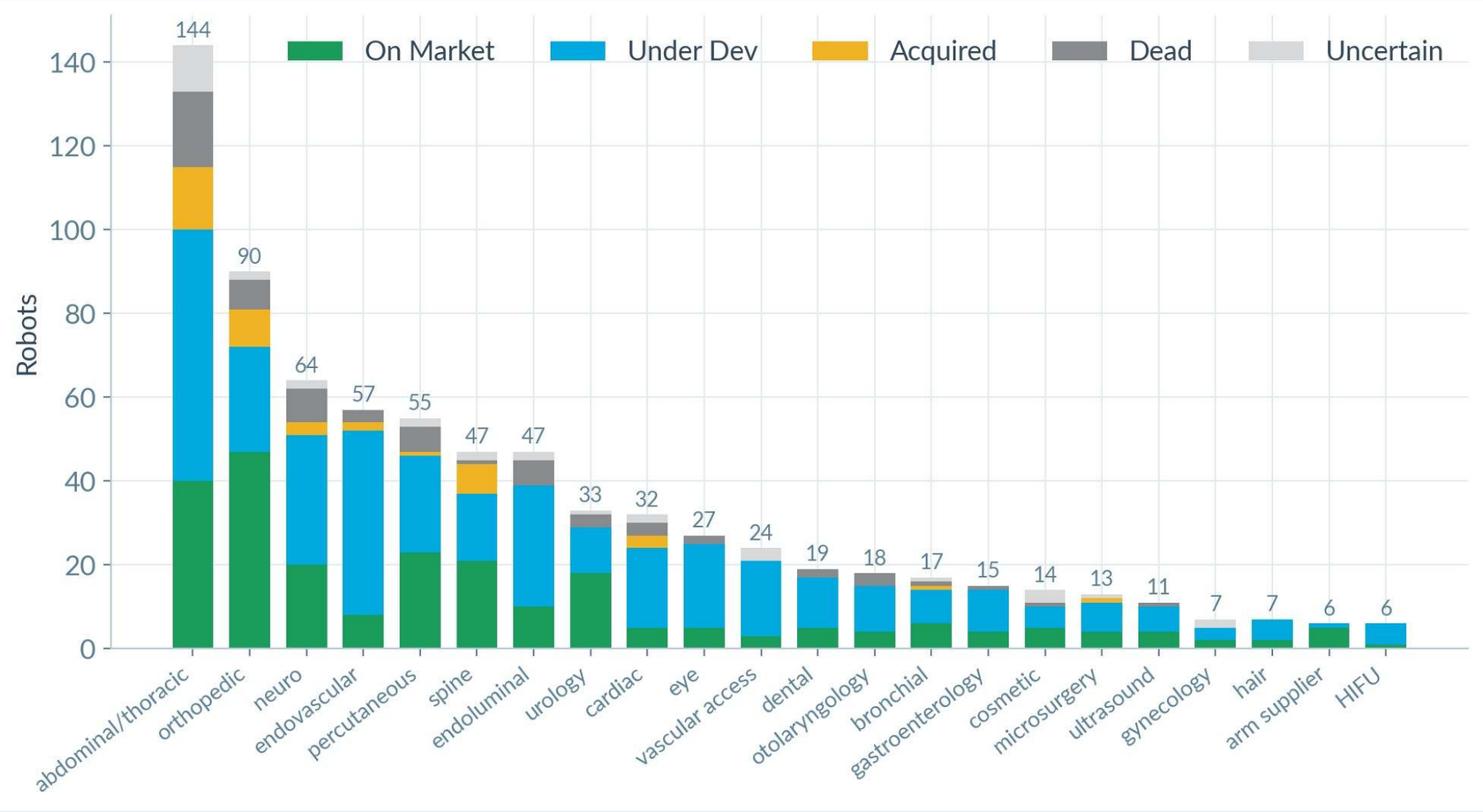


Teleoperation (488) Preplanned (176) Autonomous (82) Handheld / manual / cooperative (28)

Autonomous roughly doubled since January (37→82), partly reclassification during re-verification, partly a real trend.

Focus is still predominantly in abdominal procedures

Roster · 774 robots by focus area, colored by status



144 ABDOMINAL / THORACIC

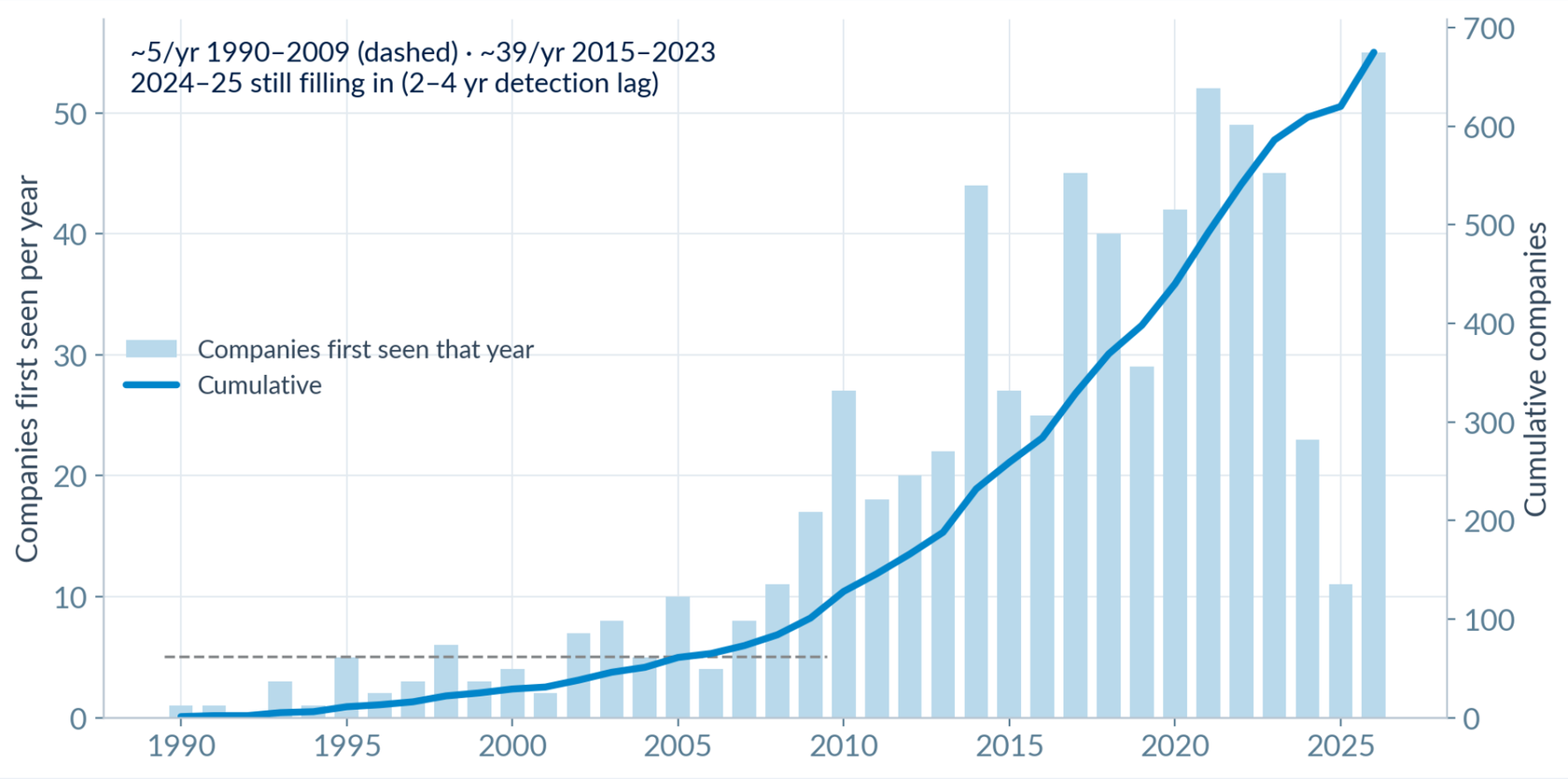
Orthopedic (90) and neuro (64) follow

A long tail of 32 focus areas

New categories since January: airway, regional anesthesia, trauma, podiatry, arm supplier

Company formation inflected around 2010

Growth · companies first seen per year, with the cumulative total



677 COMPANIES

About 5 per year before 2010

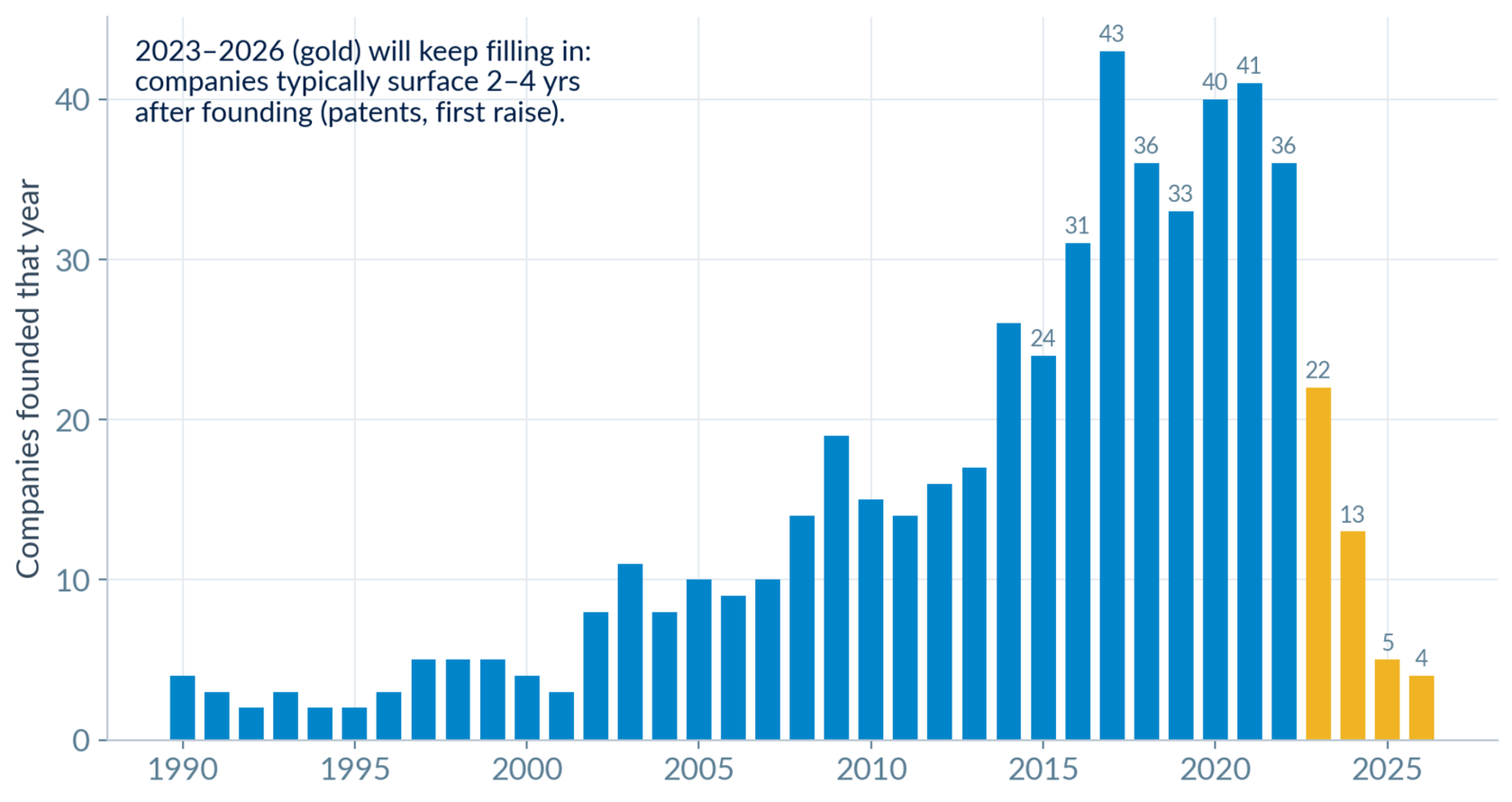
About 39 per year in 2015–2023

2024–25 look low only because companies take 2–4 years to become detectable

2026 shows a bump because of improved detection which affects first year of detection

By founding year, the field is younger than it looks

Growth · companies by founding year; the Founded column is populated for 573 of 677 companies



DETECTION LAG

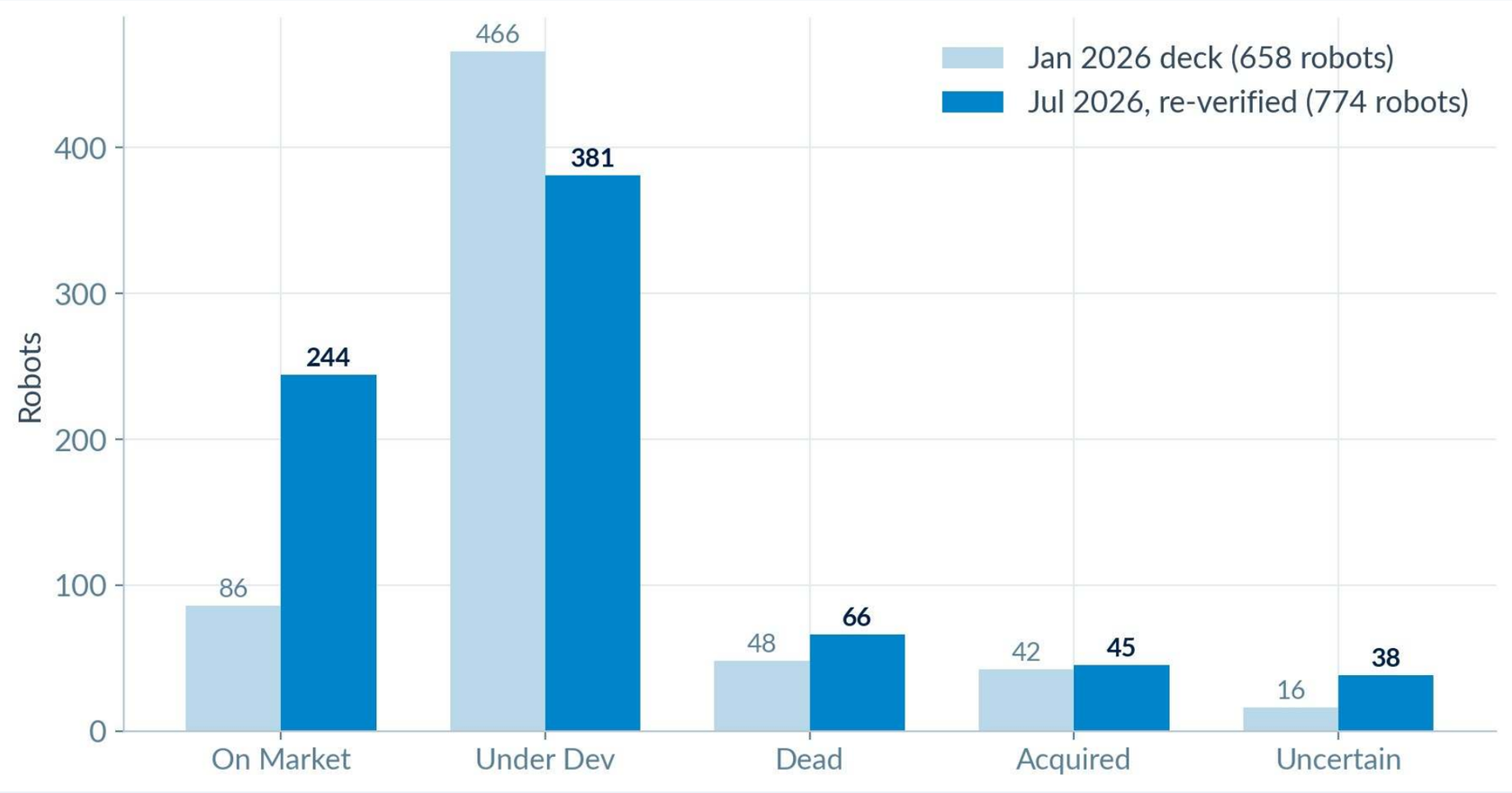
Companies typically surface 2–4 years after founding

First signals are patents and first raises, which lag incorporation

Expect the 2023–2026 bars to keep growing in future editions

The commercial wave is bigger than it looked

Growth · re-verifying every company tripled the on-market count; China's approvals were undercounted



86 → 244 ON MARKET

What changed

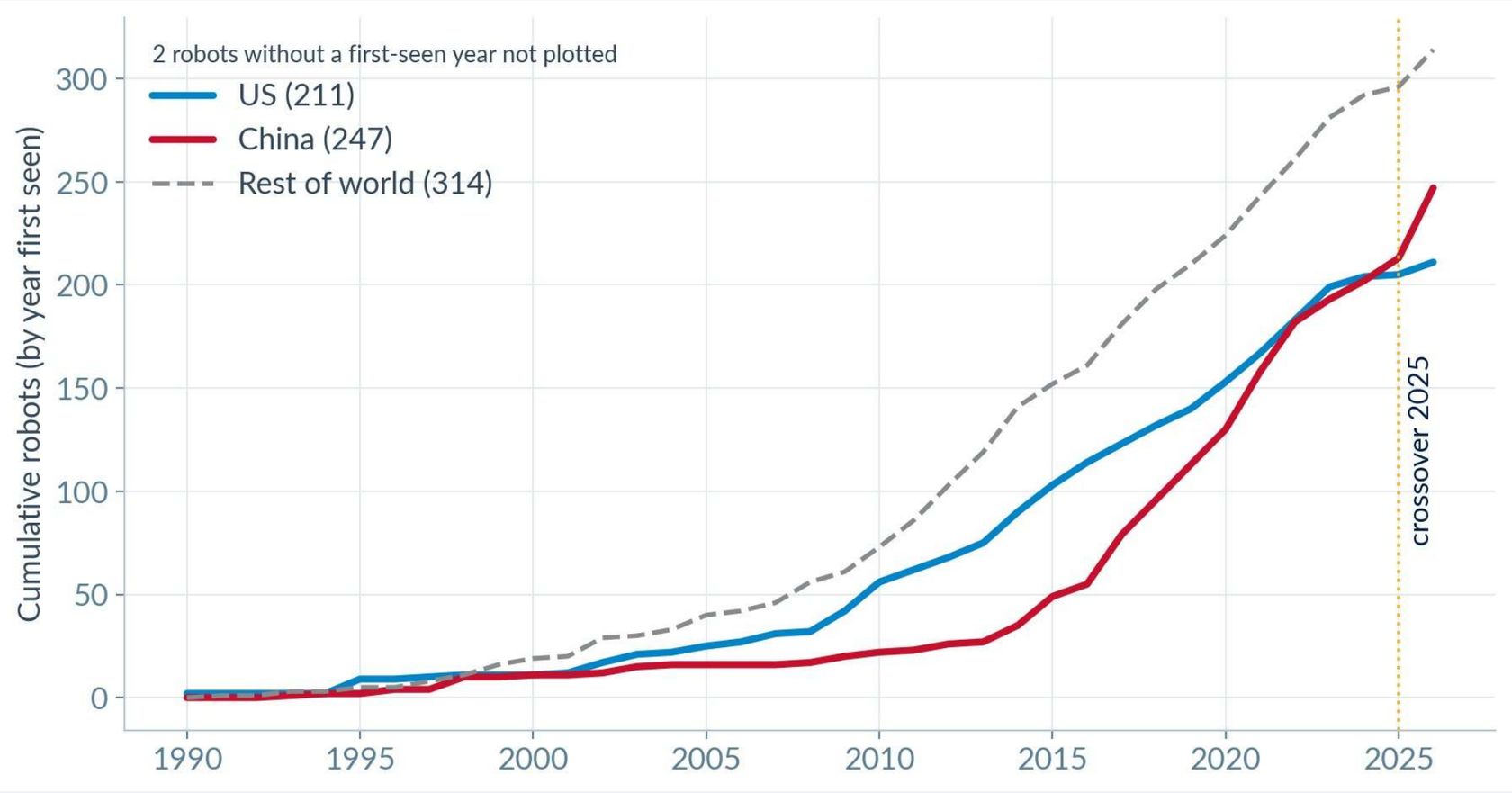
Jul 1–2 re-verification checked every row against NMPA/FDA/CE records and company disclosures

Many Chinese systems had approvals & installs that never made English-language press

Dead (48→66) and Uncertain (16→38) also grew, attrition was undercounted too

China surpassed the US in total surgical robots

Geography · cumulative robots by year first seen, China 247 vs US 211 today



247 vs 211

China vs US robots

There is detection lag in the US/China counts

Company-level counts mirror this crossover; detection lag on Chinese companies means the true gap is likely wider

Robots by country of origin

Geography · each circle is a country, area proportional to robots tracked



36 COUNTRIES

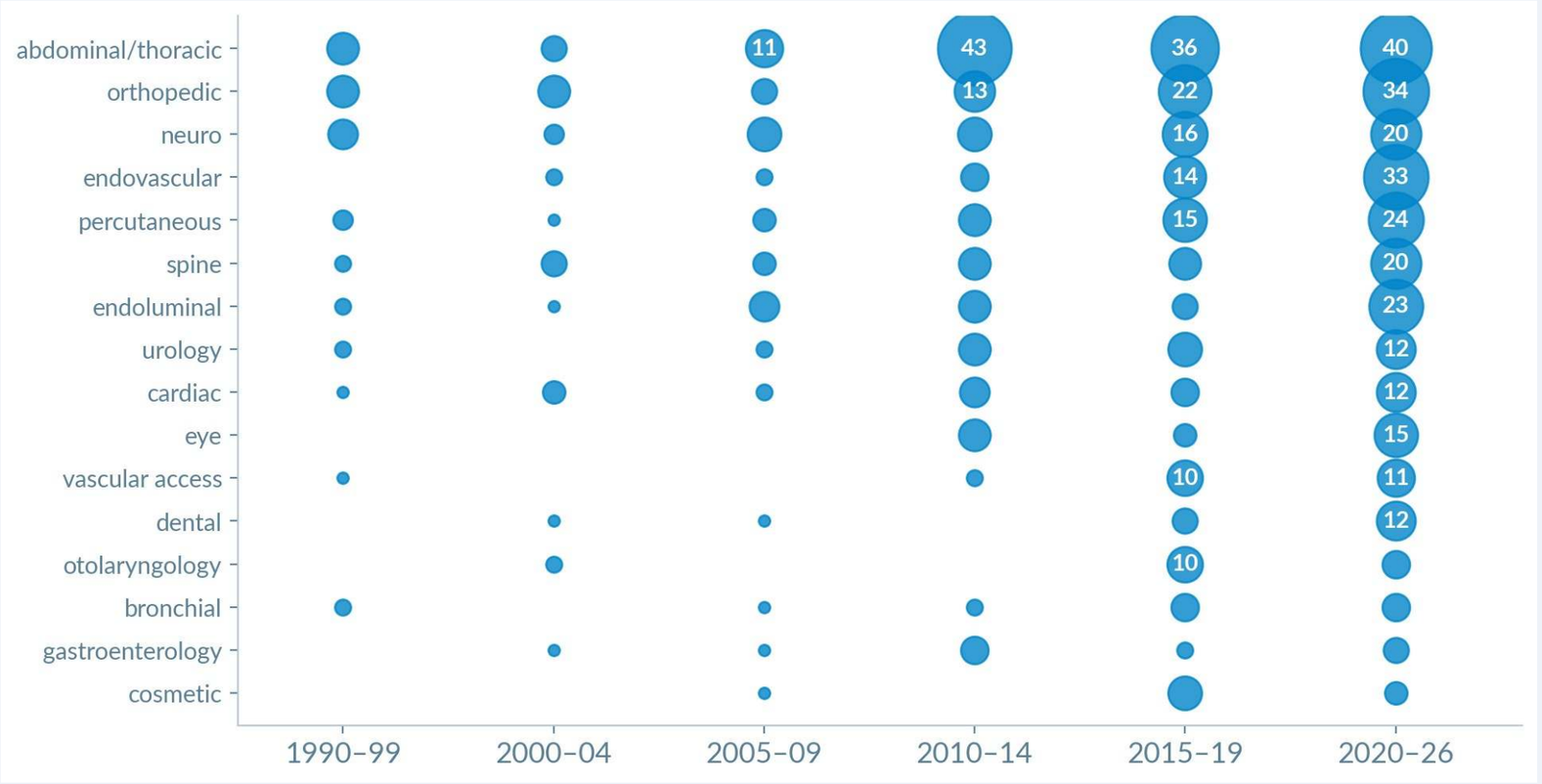
US + China = 59% of all robots

France (49), Korea (39), Germany (30) lead the tail

China's share of new robots by era: 13% through 2014, 40% in 2015–19, 43% since 2020

Global specialty robots development start

Specialty waves · Tufte-style period matrix; dot size = robots first seen in that period



Read across a row to see when each specialty became active; read down a column to see what each era produced.

China and US robots started, by specialty

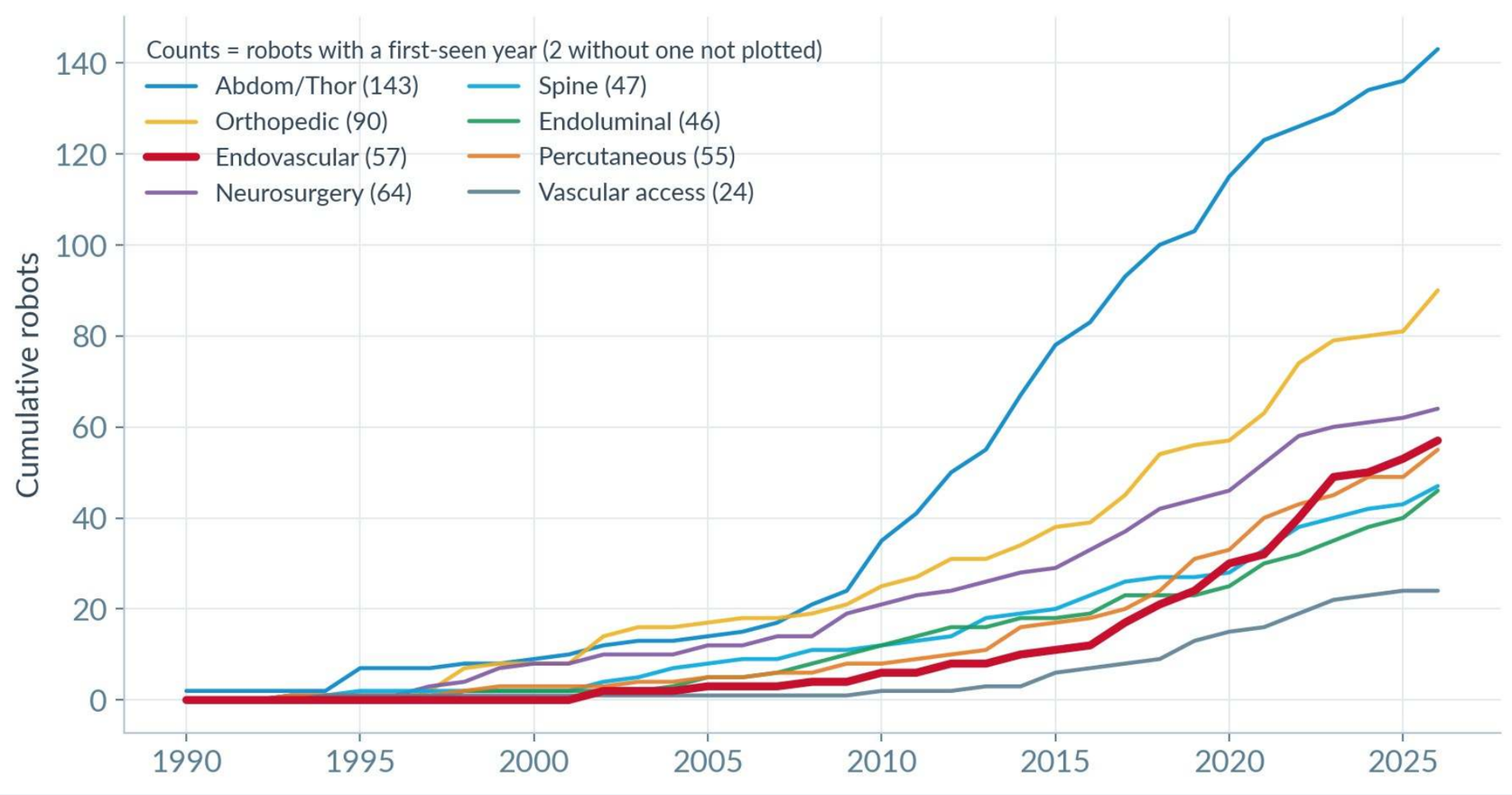
Specialty waves · side-by-side period matrices; dot size = robots first seen



The US spreads across specialties and decades; China concentrates hard in 2015–2026, especially abd/thor, endovascular, percutaneous and orthopedic.

One specialty breaks the pattern

Endovascular case · cumulative robots by specialty; one curve does not fit the pattern



57 ENDOVASCULAR ROBOTS

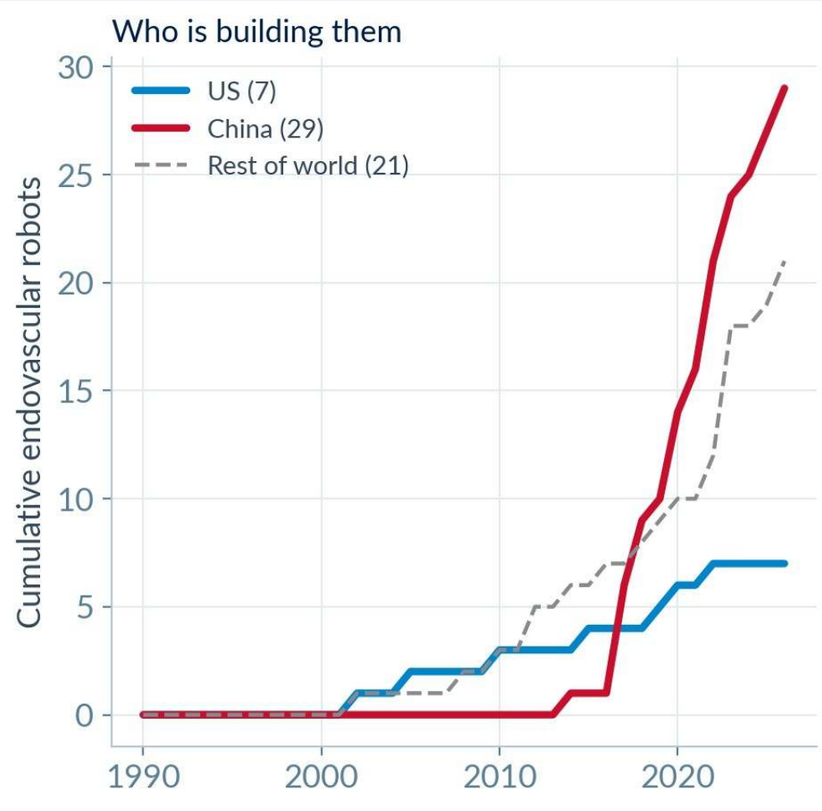
Roughly 3/4 began after 2015

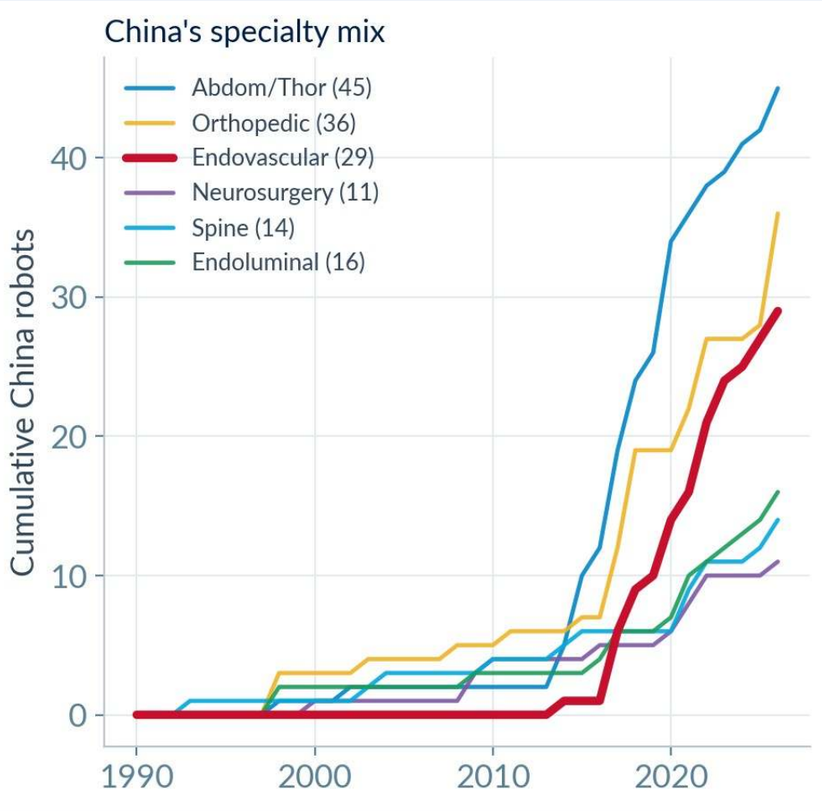
The steepest specialty curve of the last decade

China drives the spike; next slide

China accounts for the majority of the endovascular increase

Endovascular case · attribution in two views: who is building them, and China's specialty mix





CHINA'S WAVE

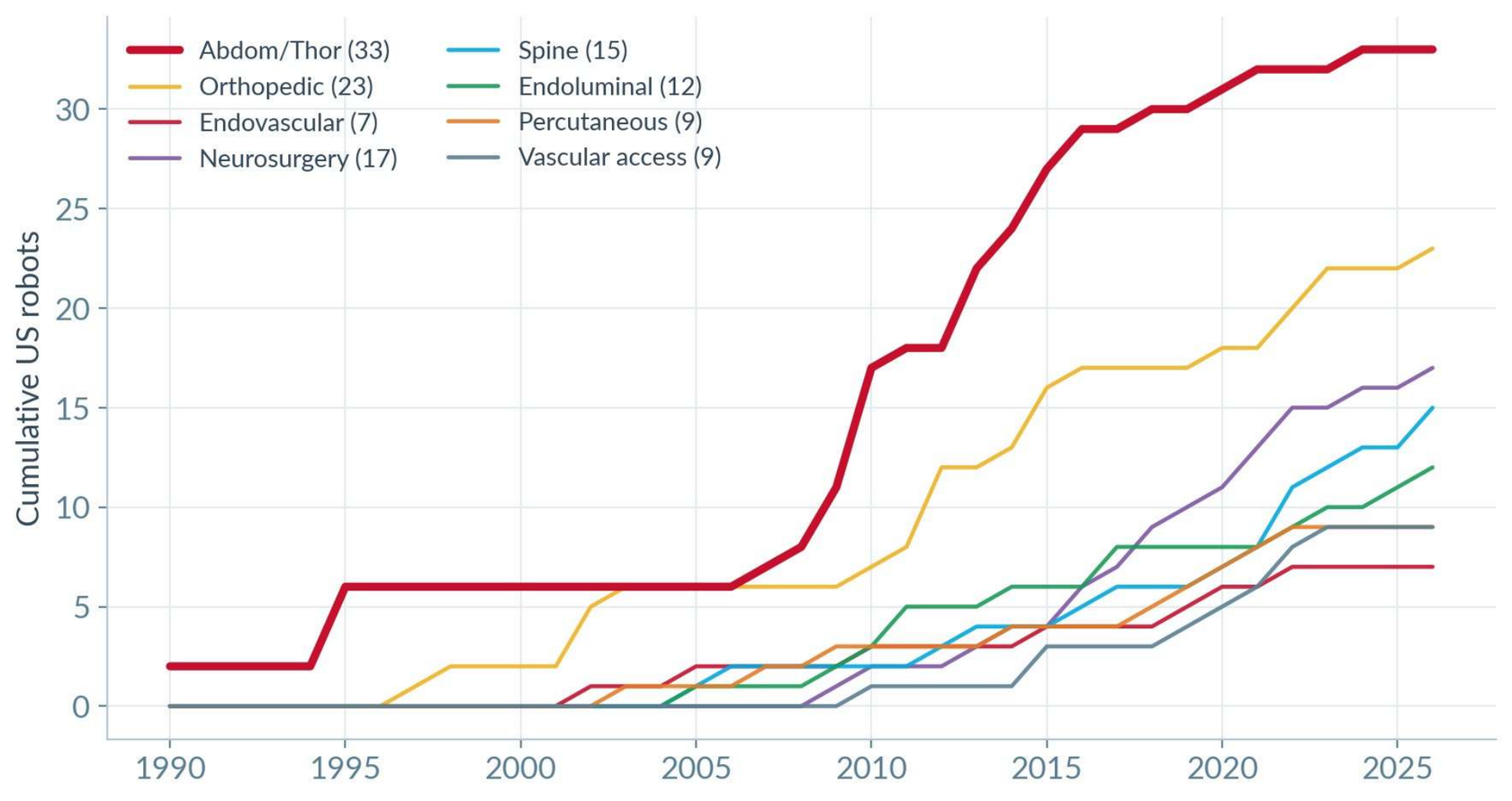
Endovascular was a US and EU niche until about 2015

China's curve bends sharply upward from 2018

Within China, endovascular is among the steepest specialty curves

US: abdominal/thoracic wave, steady orthopedic growth

Endovascular case · the US contrast: cumulative US robots by specialty



US PATTERN

Abdominal/thoracic keeps compounding despite the da Vinci incumbency

Orthopedic growth has been steady

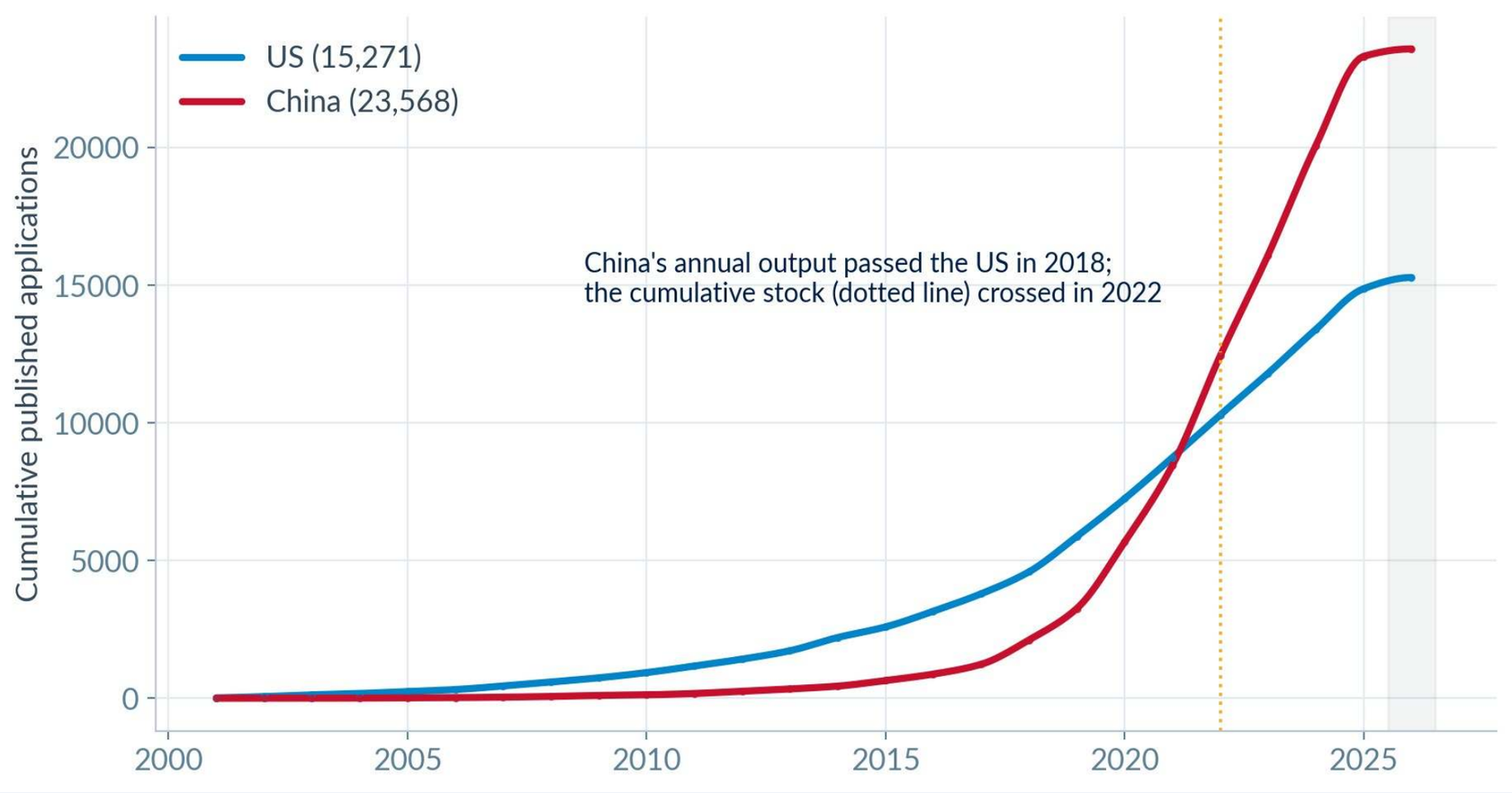
Endovascular stays a modest US category, the contrast with China is the story

Commercial surgical robotics patent trends

Four questions: how much and where · what the filings cover · who holds it · how much is genuinely new

Overall surgical robotics patent applications: US & China

Cumulative published applications, A61B34/30 incl. subgroups, like-for-like basis, refreshed 7 Jul 2026



23,568 vs 15,271

China vs US cumulative applications

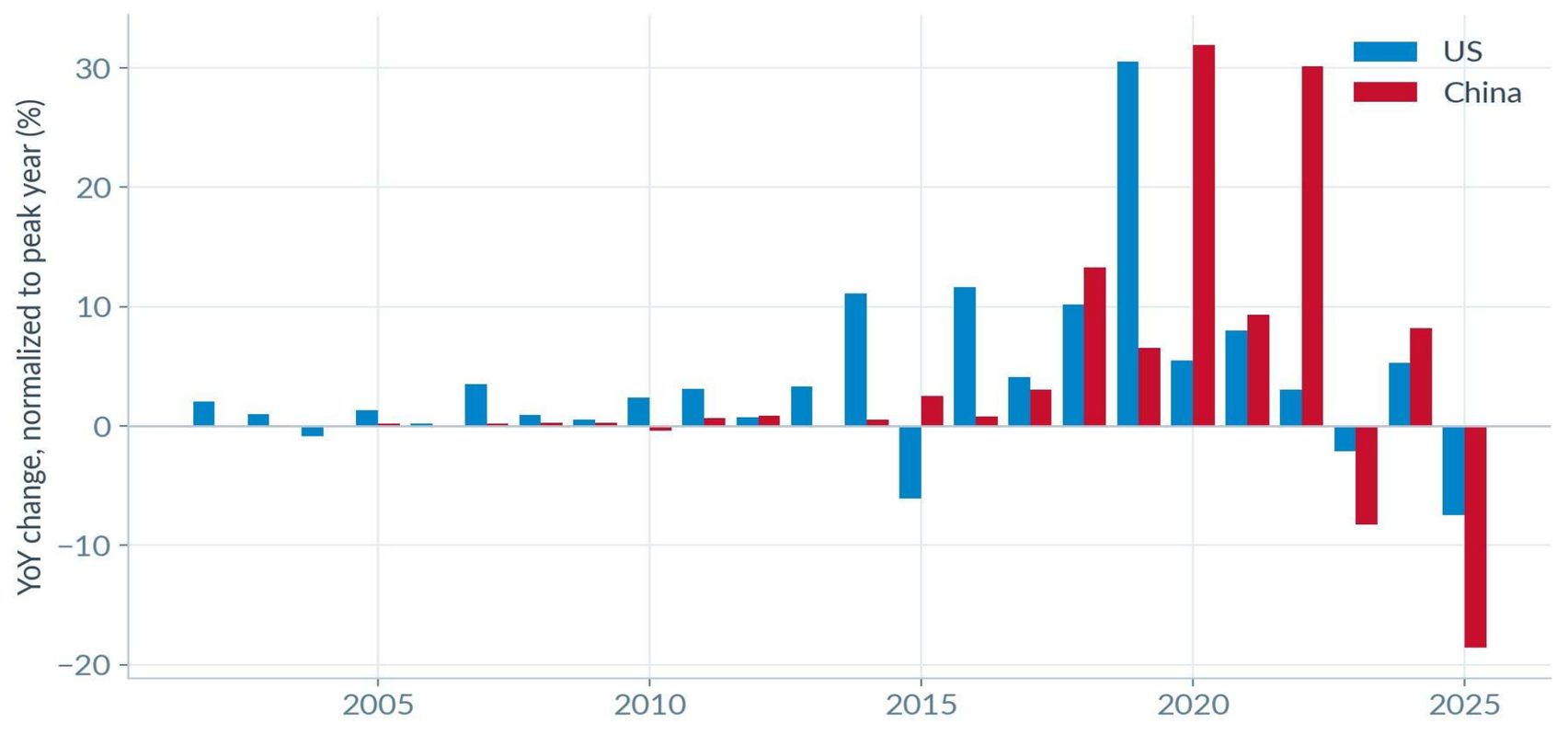
China's annual output passed the US in 2018; the cumulative stock crossed in 2022

Earlier editions mixed counting bases and overstated the gap; like-for-like, China's annual output is ~2.5x the US (2024) and its cumulative stock ~1.5x

Source: Google Patents result counts, q=(A61B34/30) country:US/CN, applications only, by publication year; collected 7 Jul 2026

US & China patents: normalized year-on-year change

(Year[n] − Year[n−1]) / peak year, normalized to each country's maximum



CHINA'S MOMENTUM

China's normalized year-on-year gain exceeded the US gain in 7 of 16 years (2010–2025); its largest jumps came in 2018–2022

Both countries decelerate after their peaks: China 2022, US 2024

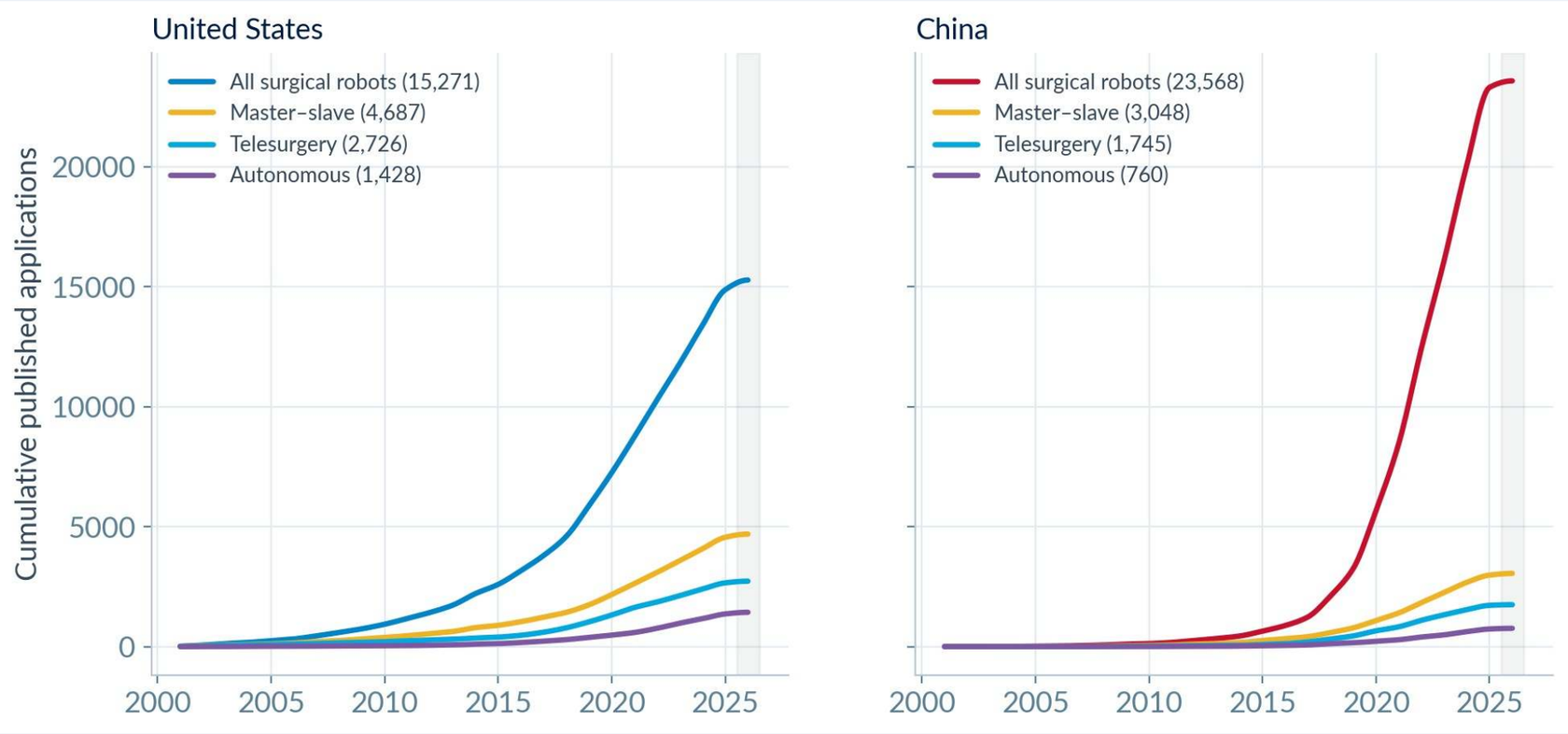
2023-2025 values are understated by the ~18-month publication lag; treat recent bars as provisional

Source: Google Patents counts, same query as the previous slide

Normalization anchors: US peak 1,595 published applications (2024); China peak 3,980 (2022). 2026 excluded (partial year).

What the filings cover: US and China by CPC subgroup

Cumulative published applications by CPC subgroup; both panels share the same vertical scale. Source: Google Patents result counts, q=(A61B34/3x), applications only, by publication year; collected 7 Jul 2026



SAME CATEGORIES, SAME SCALE

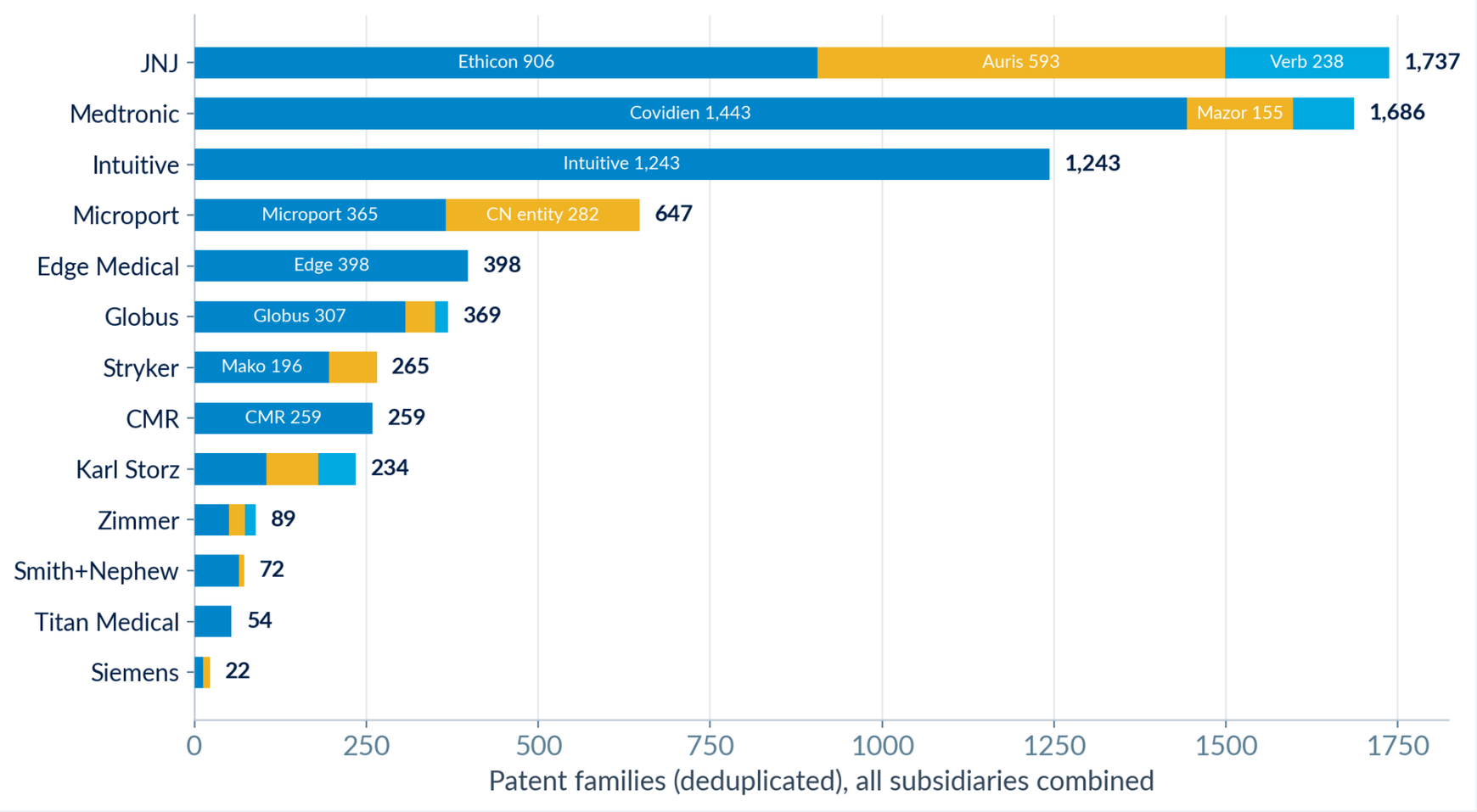
All surgical robots (34/30 incl. subgroups), master-slave, telesurgery, autonomous

China's curves accelerate the most after 2018

Source: Google Patents result counts, q=(A61B34/3x) with country and status filters, by publication year; collected 7 Jul 2026; 2026 is first-half only

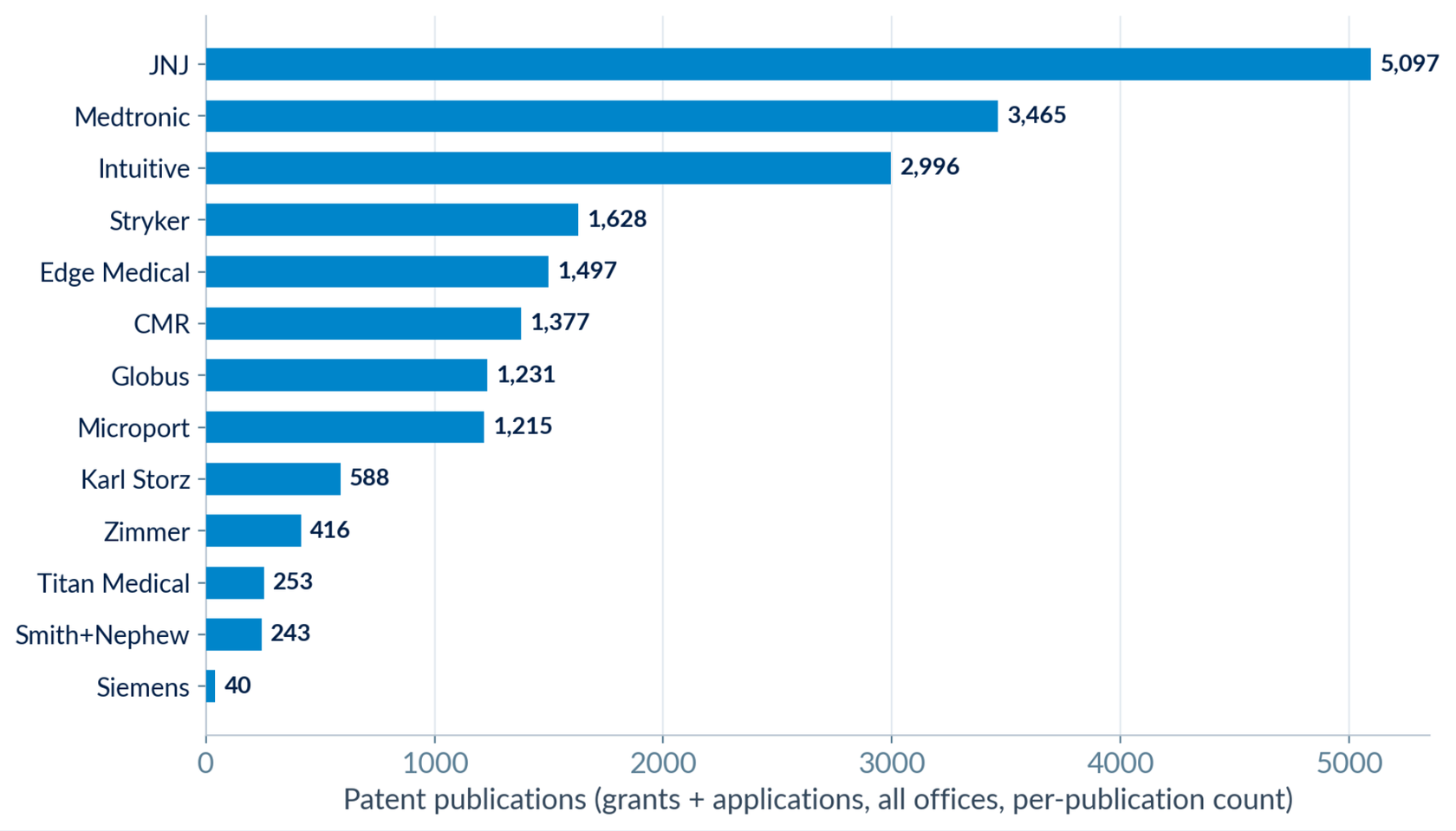
For IP, you need to count all the parts

Patent families by holder, subsidiaries stacked separately. Source: Google Patents result counts, TI/AB/CL=(surg* NEAR/3 robot*) per assignee, family-deduplicated; collected 12 Jul 2026



While there have been many patents published…

Publications (grants + applications, all offices) for the largest holders. Source: Google Patents result counts, per-publication mode; collected 12 Jul 2026



RAW VOLUME

JNJ group (Ethicon + Auris + Verb): 5,097 publications; Intuitive alone: 2,996

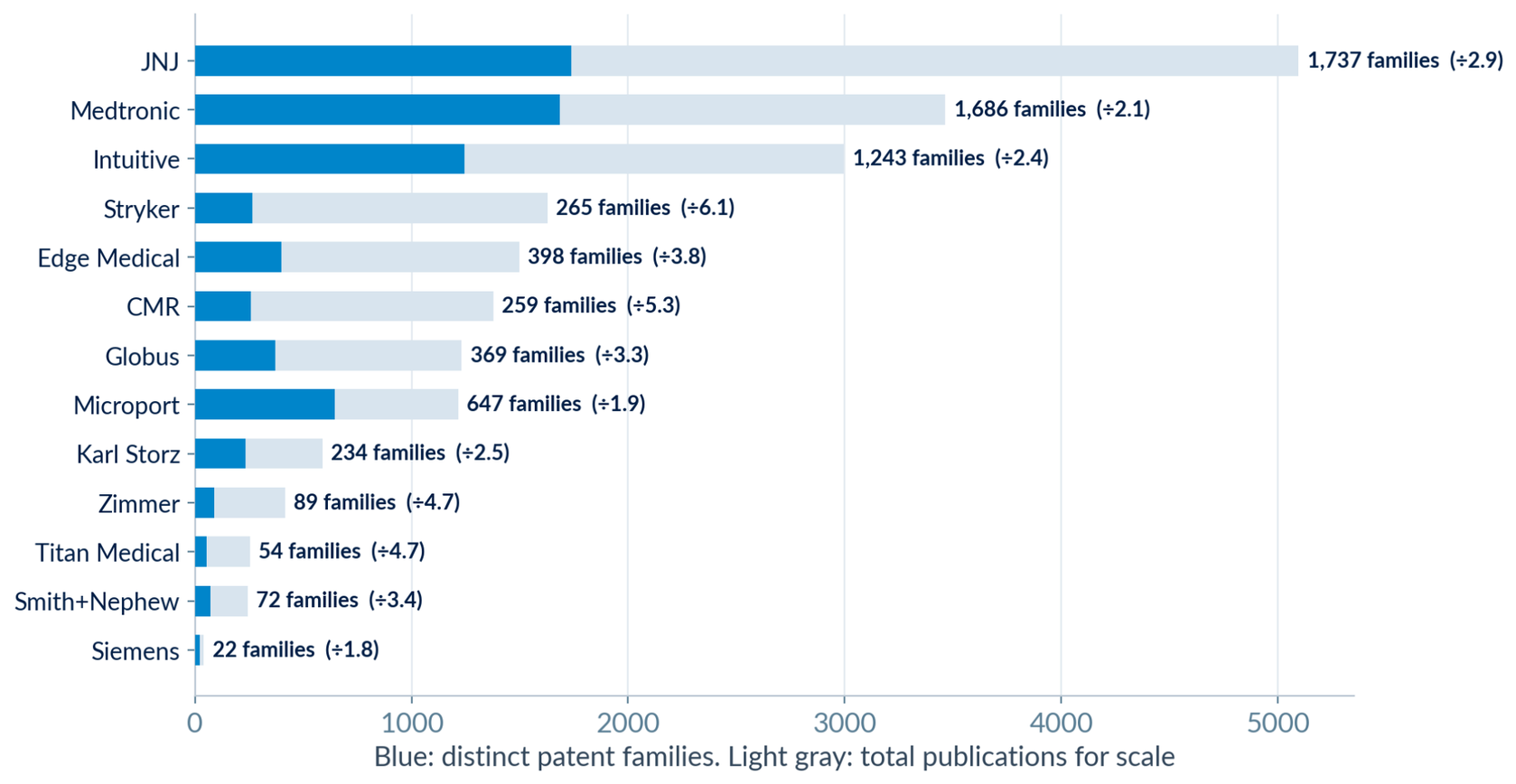
Publication counts include every office and every continuation

Volume alone overstates novelty, next slide

Source: Google Patents, collected 12 Jul 2026

...many are extensions of existing IP

Distinct patent families vs total publications, same query. Source: Google Patents result counts, family-deduplicated vs per-publication; collected 12 Jul 2026



FAMILY-LEVEL VIEW

Families run about 2x to 6x below publication counts (Stryker 6.1x, CMR 5.3x, Intuitive 2.4x)

JNJ group: 1,737 families across Ethicon, Auris, and Verb

For diligence, count families and claims, not publications

Source: Google Patents, collected 12 Jul 2026

Examples of surgical robots in different specialties

Pictures shown not remotely comprehensive: intentionally chosen to provide historical perspective
Note that DEAD classification refers to the robot, not necessarily the company

Abdominal / Thoracic

Representative examples



INTUITIVE SURGICAL



computermoti

CMR SURGICAL



EDGE MEDICAL 精锋医疗



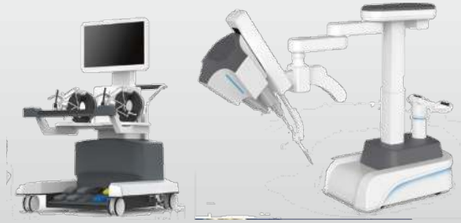




Medicaroid



ENDOQUEST ROBOTICS



TITAN MEDICAL

endoVia

ASENSUS SURGICAL



Abdominal / Thoracic

144 robots from 129 companies

34
ON MARKET

Agibot Medtech
AKTORmed
Applied Dexterity
Asahi Surgical Robotics
CMR Surgical
Cornerstone Robotics
Dextérité Surgical
Distalmotion
Edge Medical Robotics
FreeHand Surgical
Genesis MedTech Group
HIWIN Technologies Corp.
Huger Medical Robotics
HumanTouch Surgical
Innolcon / 以诺康
Intuitive Surgical
JNJ MedTech
Levita Magnetics
LivsMed
Meere Company
Merai Newage
Moon Surgical
Parallel Robotics LLC
Precision Robotics
Ronovo Surgical
Sagebot / 思哲睿
Sina Robotics and...
SS Innovations
Tuebingen Scientific...
Tuodao Medical
United Imaging Surgical
Virtual Incision
Weigao Group / 威高集团
Wise King / 唯精医疗

53
UNDER DEV

Accrea Medical Robotics
All-Star Medical
Aopeng Medical
Apeiron Surgical
Articulus Surgical
ARTS GmbH
Asvel OÜ
B Braun New Ventures
Baining Yingchuang...
Borns Medical Robotics
Canady Robotics
DEKA Research &...
Denovo Bioinnovations
Doosan Robotics
Eabmed Science and
Erbe Elektromedizin
GmbH
Fulbright Medical
Gamania Digital...
IMRobonix
Jinshan Science &
Kirubotics Surgical...
Macheon AI
Maituo Medical
Mars Medical Robot
Meike Electronic...
Micron Shiga
MindAdvance Medical
Mindray / 迈瑞医疗
Nikkotech
NISI
Noahtron Intelligence...
Novatimes Intelligent
Nuclear Medical Robot
Olympus Corporation
Optosurgical, LLC
Orvue Surgical A/S
Panther Healthcare
Rainbow Robotics
Revolve Surgical
Rob Surgical Systems...
Ruida Medical Devices
Scichip Robotics
Shengzhi Medical
Shukra Robotics Pvt.
Steerable Instruments...
Symphera GmbH
Titan Medical
ValueBioTech S.R.L.
Vicente Medical Robot
Viestar
VinSurgical
XSurgical Robotics
Yinuoda Medical

14
ACQUIRED

Asensus Surgical
Auris Health / JNJ
Computer Motion
Covidien
Endocontrol SA
Gomtec / ABB
Great Belief...
Human Xtensions
JNJ Ethicon...
Medical Surgery
SOFAR S.p.A.
TransEnterix
Tuge Healthcare
Verb Surgical

11
UNCERTAIN

Assisted Surgical
B Braun
Ekymed
EndoRobotics, Inc.
Kezhixing Robot
Medsys SA
Modular Surgical
Movasu
Shuchuang Robotics
SRI International
Virtu Surgical

17
DEAD

Agile EndoSurgery
American GNC
Corporation
Avatera Medical
AVRA Medical Robotics
Bio-Medical Engineering
Human Robotics S.A. de...
Invivo Robotics
Lapara Surgical
LG Electronics
MDA
Omniboros
Panasonic Holdings
Roboscopy Medical Ltd.
Samsung Electronics
Signature Robot Ltd
SPI Surgical, Inc.
Vicarious Surgical Inc.

Orthopedics

Representative examples



MAQUET
GETINGE GROUP



stryker
Orthopaedics

MAKO
SURGICAL CORP



ROBODOC
Curexo Technology Corporation

THINK
SURGICAL



ZIMMER BIOMET
medtech

DePuy Synthes
THE ORTHOPAEDICS COMPANY OF Johnson&Johnson



VELYS™
Digital Surgery



stryker
Stanmore Implants

acrobot
Precision Surgical Systems

AOT





smith&nephew

BLUE BELT TECHNOLOGIES, INC



Orthopedics

90 robots from 79 companies

38
ON MARKET

Aimooe / 艾目易
Airs Inc
AK Medical
Amplitude Surgical
Beijing Visual Medtech
Chunli / 春立医疗
Corin Apollo
Curexo / 큐렉소
DePuy Synthes
DroidSurg Medical
Enovis
FUTURTEC Medical
Globus Medical
Gusheng Yuanhua Robot
Howmedica Osteonics
Hurwa / 和华瑞博
Jianjia Medical
Kuangrui Intelligent
Lancet Robotics
Longwood Valley...
McGinley Orthopedics
Medos
MedTech SA
Merai Newage
Meril Healthcare
MicroPort MedBot
Naton Medical Robotics
Rossum Robot
Santa Medical
Smith & Nephew plc
Stryker
Surgify Medical
THINK Surgical
Tinavi Medical
Tuodao Medical
Unik Orthopedics, Inc.
Weigao Group
Yidian Lingdong

23
UNDER DEV

Aopeng Medical
ArthroLase
Bozhi Bio
Chain Orthopedics, LLC
Connecteve
Convergence Medical
Dading Medical
Duluth Medical
Extremis Robotics
Ganymed Robotics
Great Robotics
Hangwei
Interventional Systems
Jieduo
Jimuu Intelligent Tech
Kyniska Robotics
Laishbo Medical
Linkwah Technologies
Medical Robotics i...
Robossis, Inc.
Tianxing Medical Co.
Xingyun Guiyi...
Xinkangyuan...

9
ACQUIRED

Blue Belt Technologies
KAT Robotics
LinkX Robotics
Mako Surgical Corp.
Monogram Technologies
OMNIlife Science
OrthoSpin Ltd.
Orthotaxy SAS
Ostesys

2
UNCERTAIN

Seva Robotics
Shukang Medical

7
DEAD

Acrobot / Stanmore
Advanced Osteotomy...
iin medical
Longhui Medical
ortoMaquet GmbH
Revision Robotics...
Sagitta Robotics

Spine

Representative examples

Medtronic

Inspiring the Art of Surgery
Mazor Robotics





GLOBUS MEDICAL

LEM surgical



GLOBUS MEDICAL

KB MEDICAL



CUREXO



INTERVENTIONAL SYSTEMS



Spine

47 robots from 46 companies

20
ON MARKET

ATEC
Bangjiexing Medical
Curexo / 큐렉소
Cyber Surgery ALAYA
Dabo Medical / 大博医疗
eCential Robotics
LEM Surgical
MedTech SA
Medtronic
Meyer Optoelectronic
MinMaxMedical
Perlove Medical / 普爱医疗
Point Robotics MedTech
Remex Medical Corp.
Sino-Europe MicroRobotics
SMAIO SA
SMTP Technology Co., Ltd.
Sylorus Robotics
Xinjunte Smart Medical...
Zhuzheng Robot / 铸正机器人

16
UNDER DEV

8i Robotics
Automated Intelligent...
Carevature Robotics
Chunfeng Huayu Smart Medical
Deneb Medical
Dianhe Medical Technology
FoxEyes Corporation
Kato Medical / KatoMed
Koellner Medical
Linyan Medical
Maestro Surgical Robotics
Robotron Surgical
Sanyou Medical
Surgitec Robotics
Vista Robotics, Inc.
Weigao Group / 威高集团

7
ACQUIRED

Cardan Robotics
Excelsius Surgical
Fusion Robotics / Accelus
KB Medical
Mazor Robotics
Mobius Imaging / Stryker
Relievant Medsystems

2
UNCERTAIN

MI4Spine, LLC
Teslake, Inc.

1
DEAD

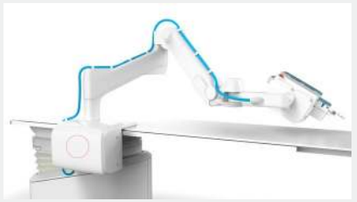
Rubicon Spine, LLC

Endovascular

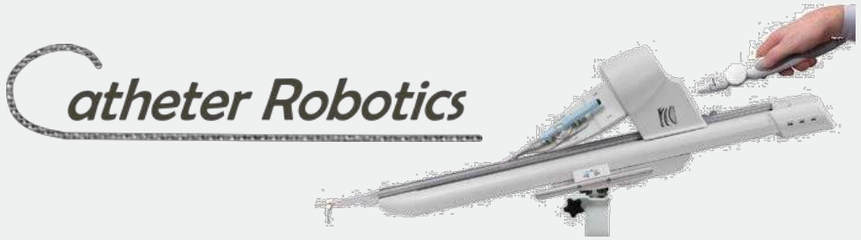
Representative examples



Corindus
A Siemens Healthineers Company



Robocath



catheter Robotics



microbot medical



hansen MEDICAL



STEREOTAXIS

aeon scientific



Endovascular

57 robots from 56 companies

8
ON MARKET

Aibo Medical Robot
Artiria Medical
Inovatyvi Medicina
Microbot Medical
MicroPort MedBot
UnionStrong
WeMed / 唯迈医疗
Yidu Hebei Robot Technology

43
UNDER DEV

Angio8
Aopeng Medical
Apeiron Surgical
Basecamp Vascular
Biomedical Research Center
Biotronik AG
Calabrian High Tech
Curie Interventional...
Dashi Technology
Dinova Neuroscience
EndoWays / Ascendum
Flux Robotics
Great Robotics / 歌锐科技
Hanglok-Tech
Huada Zhizao Yunying Medical
Huaxi Jingchuang Medical
InterMag
Jieruo Medical Technology
Jingge Medical Technology
Koswire / 코스와이어
Langbokang Medical
Langya Hechuang Medical
Magellan Biomedical Inc.
MagnebotiX
MiraCure Co., Ltd.
Nano4Imaging
New Vascular
Nimble Surgical
Perazah Inc. / 페라자
RainMed Medical
Raysight Medical
Remedy Robotics
ROBGENIX MEDICAL PTE. LTD.
Route 92 Medical
Shenganjie Medical
Shengyi Technology
Terumo Corporation
Vital Biomedical
White Bear Medical LLC
Wirescience Medical
Yindatong / 盈达通
Zero Radiation Intervention
Zhongke Hongtai Medical

2
ACQUIRED

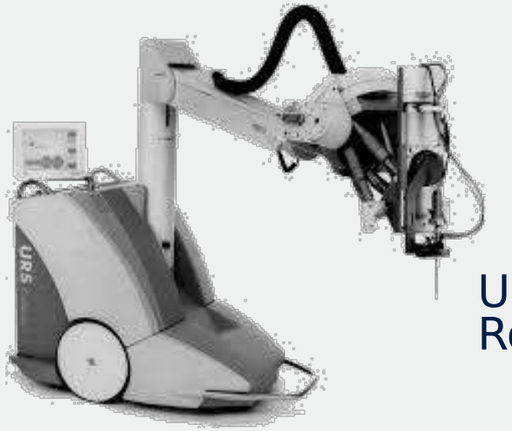
Corindus
RoboCath

3
DEAD

Aeon Scientific
Catheter Precision, Inc.
Hansen Medical / JNJ

Neurosurgery

Representative examples



Universal
Robot Systems

Axilum Robotics
Assisting you in technical procedures





medtech



ELEKTA





IMRIS
DEERFIELD IMAGING

AiM MEDICAL ROBOTICS



RENISHAW
apply innovation™



OrbSurgical Ltd.

Neurosurgery

64 robots from 61 companies

19
ON MARKET

ANT Neuro
Ariemedi / 艾瑞迈迪
Axilum Robotics
B Braun
Brain Navi…
Brainlab
Elekta AB
HOZ Medical / 华志微创
IMRIS Imaging
Insightec
Koh Young Technology
MedTech SA
NovaSignal
Rapid Medical
Remebot / 柏惠维康
Renishaw plc
Sinovation Medical
Vas Medical
Zeta Surgical

30
UNDER DEV

AiM Medical Robotics
Bionaut Labs
ClearPoint Neuro
Epica International
Esspen GmbH
Ezmedibot
Genlight MedTech
HROCT LITT Medtech
IMSystem / 아임시스템
Inetec Nero
Magnendo
Nanoflex Robotics
Navigantis
Neuralink
neuro42
Neurosputnik
NeuroXess
OrbSurgical Ltd.
Panda Surgical
Project 1985, LLC
Robeauté
Rogue Research
RONNA Medical
SceneRay / 景昱医疗
Sibo Huiyi Tech
Siemens Healthineers…
Telos Health
UNandUP
Weigao Group
XCath

3
ACQUIRED

ArteDrone
MDA
Medineering / Brainlab

2
UNCERTAIN

Signate Life Sciences
TheraSonic

7
DEAD

Centauri Robotic…
CranioMedix Inc.
Euphrates Vascular
ISIS
MDA Space Ltd.
NeuroArm / SYMBIS
Synaptive Medical

GI Endoluminal

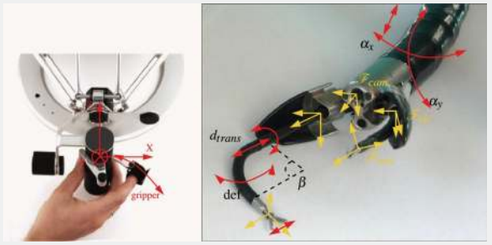
Representative examples

OLYMPUS





Endotics



STORZ
KARL STORZ — ENDOSKOPE

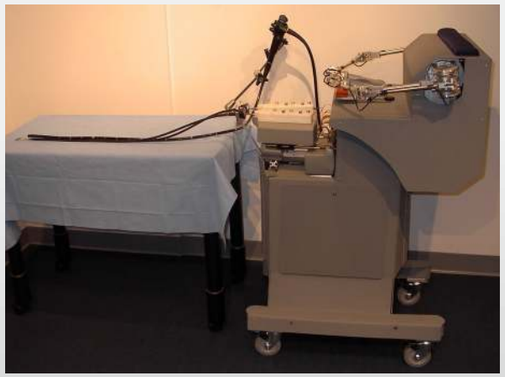
NeoGuide
Control your path.







endoVia



EndoMaster



TRITON
MEDICAL ROBOTICS

ENDOQUEST
ROBOTICS



Endoluminal

47 robots from 45 companies

9
ON MARKET

Ankon Technologies
EndoRobotics Co., Ltd.
Endotics System
G.I. View Ltd
JIFU Medical
Jinshan Science & Technology
MedInTech
Neptune Medical
Nitinotes Surgical

29
UNDER DEV

Agilis Robotics
Aohua Endoscopy
Aopeng Medical
Atlas Endoscopy
Axtes / Aiketes
Baifu
Channel Robotics
EndoQuest Robotics
EndoR Surgical System
Ergami Endoscopy
Fall River Laboratories
Huiyue Medical
HumanTouch Surgical
INEN Robotics
InterMag
Inventor Technology
Keypoint Surgical
NGScopes
NISI
Radius Dynamics GmbH
Serpenta
Softech Robotics
Swan EndoSurgical, Inc.
Ultimage Health Technology
Vivo Surgical
W Endoluminal Robotics
Weihai Kefan Medical Device
Yunlijing'an Technology
Yuying Medical International

2
UNCERTAIN

Boston Scientific
Fractyl Health

5
DEAD

Invendo Medical / Ambu
Medrobotics Corporation
MichelAngelo RobotScope
Motus GI Medical
NeoGuide Systems

Lung Endoluminal

Representative examples

ion



INTUITIVE



GALAXY

Noah Medical

AURIS

Johnson&Johnson MedTech



MONARCH™ QUEST

SPIRO Robotics



Lung Endoluminal / Bronchial

20 robots from 20 companies

7
ON MARKET

Edge Medical Robotics
Intuitive Surgical
JNJ MedTech - Monarch
Langkai Medical
MicroPort MedBot
Noah Medical
Spiro Robotics

10
UNDER DEV

Anteeo Surgical
Apeio Health Co., Ltd.
Apreo Health
Broncus Medical / 堃博医疗
Hanglok-Tech
LungHealth Medtech
Magnisity
SyncRobotix, Inc.
Vine Medical, Inc.
Xuzhou Yintu Medical...

1
ACQUIRED

Auris Health

1
UNCERTAIN

Sanovas, Inc.

1
DEAD

Borns Medical Robotics

Urology

Representative examples



edap tms

Bringing New Horizons to Therapy



VIRTUOSO SURGICAL



HistoSonics®

Clinical trials for kidney



PROCEPT BioRobotics



ROBOTICS



biobot

Urology

33 robots from 28 companies

16
ON MARKET

BestMT / 拜斯特医疗科技
Biobot Surgical
ELMED Medical Systems
FocalTherics / EDAP TMS
Innomedicus / ARTEMIS
Intuitive Surgical
Jiehang Robot / 上海介航机器人
JNJ MedTech - Monarch
Medicaroid Corporation
Medtronic
MicroPort MedBot
PROCEPT BioRobotics
Profound Medical
ROEN Surgical / 로엔서지컬
Sterlab
Surgerii Robotics

9
UNDER DEV

Andromeda Surgical
Healinno Tech
Insight Medbotics
LARC Robotics
Needleye Robotics Srl
Promaxo, Inc.
Richard Wolf GmbH
Sagebot / 思哲睿
Virtuoso Surgical

1
UNCERTAIN

Continuum Therapeutics

2
DEAD

Comau SpA
Polymer Robotics Inc.

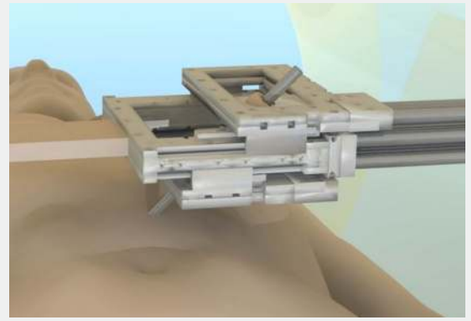
Percutaneous

Representative examples



Perfint

New Directions in Interventional Oncology



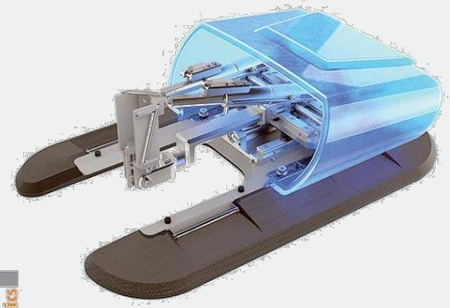
GUIDABOT, LLC

SYNAPTIVE MEDICAL





MENDAERA



XACT Robotics



Percutaneous

55 robots from 52 companies

21
ON MARKET

CAScination
Ganaivi Medical
Hanglok-Tech
HistoSonics
Insight Medbotics
Insightec
Interventional Systems
LifeTech-Quantum Medical
Longdianjing Intelligent
MASMEC Biomed
Mendaera
NDR Medical Technology
Perfint Healthcare Pvt Ltd
Praxis Holding LLC
Quantum Surgical
Ruitouch Technology
Soteria Medical
Tuodao Medical
Weide Precision Medical
Zhenjiankang Medical
Zhongrui Funing / 中瑞福宁

23
UNDER DEV

AIR Surgical
AIRS Medical
Aoda Zhisheng Medical
Apodibot Medical
Baird Medical
BEC / Guidoo
Beijing Yiru Jingzhi Medical
Cosmic M.E.
Cure-in
DEMCON Medical Robotics
Edda Technology
FlexPerc Medical
Hengrui Hongyuan Medical
Libang Surgical Technologies
LUKE Robotics
Medical Devices Corner / MDC
Miaoweishi Technology
Nawei Medical / 奈维医疗
Puncture Robotic / 磅策医疗
Shengdankang Medical
Xianyang Robot Co.
Yida Health Medical...
Yingtai Li'ankang

1
ACQUIRED

Innomedic / Synthes / JNJ

1
UNCERTAIN

RobotX Medical

6
DEAD

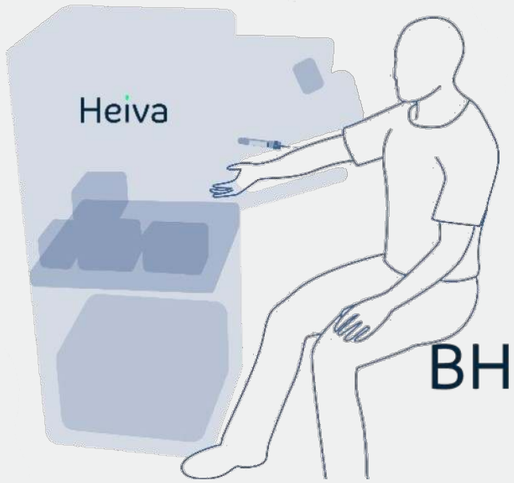
GuidaBot LLC
Robopsy
Seiko Epson
Starpax Biopharma
XACT Robotics Ltd.
Yikangtai Medical Technology

Vascular Access

Representative examples



VITESTRO



BHealthCare



VEEBOT SYSTEMS



Vascular Access

24 robots from 23 companies

3
ON MARKET

Kairui Medical / 凯瑞医疗
MagicNurse / 迈纳士
Vitestro

17
UNDER DEV

AutonomUS Medical Technologies
BHealthCare
Bracco Imaging
Cobionix
Creare LLC
Faction Imaging
Fresenius
Fuxi Jiuzhen Intelligent Technology
Grifols Worldwide Operations Limited
Hyperion Surgical
Nigale Biomedical / 四川南格尔生物科技
Obvius Robotics
Puncture Robotic / 磅策医疗
Takeda Pharmaceutical / 武田薬品
Triton Systems
VascuLogic, LLC
X9, Inc.

3
UNCERTAIN

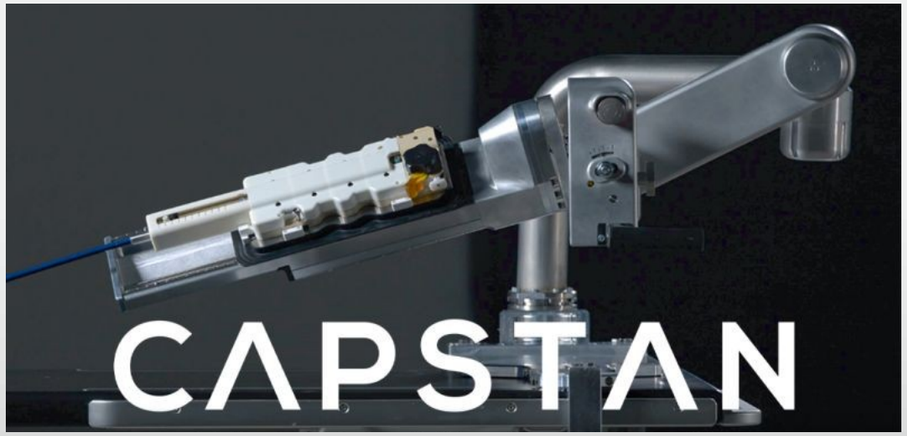
Gliatech SL
Sheba Impact
Veebot Systems, Inc.

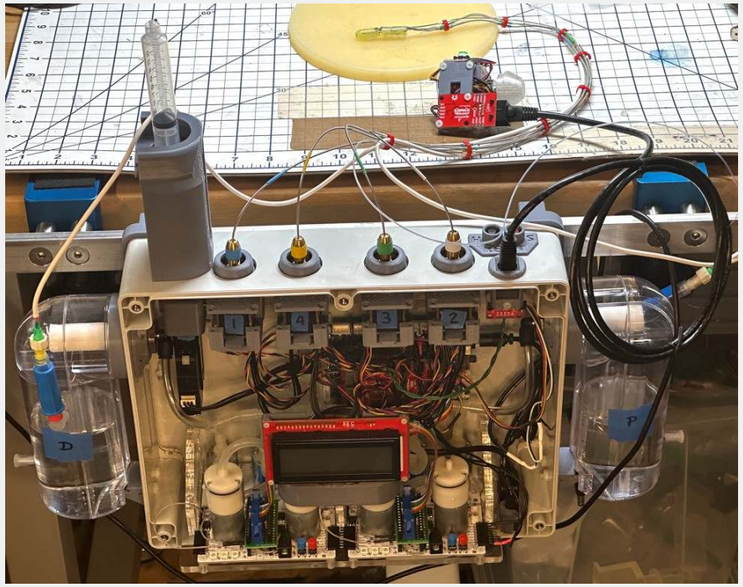
Cardiac

Representative examples



CARANX MEDICAL





HeartLander™ Surgical Inc

Cardiac

32 robots from 32 companies

5
ON MARKET

Cardiawave
LN Robotics
Mayo Cardiomagnetics
Stereotaxis, Inc.
Transmural Systems

19
UNDER DEV

Biosense Webster
Capstan Medical
Conform Medical
Denuo Electrophysiology…
DigiSurge
Edwards Lifesciences
EndoSurge
Indian Wells Medical
Jenscare Scientific
Laza Medical
Neuro-Kinesis Corp.
Peijia Medical / 沛嘉医疗
Pnemed Technology Limited
ROB'E GmbH
Shifamed Holdings
Surlogic
Symphony Robotics
UNandUP
Zhicheng Medical / 智程医疗

3
ACQUIRED

Caranx Medical
CyberHeart / Varian
VytronUS, Inc.

2
UNCERTAIN

CardiaTouch Control Systems
HeartLander

3
DEAD

Hansen Medical / JNJ
Moray Medical
St. Jude Robotic Catheter

Microsurgery

Representative examples



MMI®



SONY



GALEN ROBOTICS



microsure
robot assisted microsurgery

Microsurgery

13 robots from 13 companies

4
ON MARKET

BHS Technologies
Lydus Medical
Medical Microinstruments
Microsure

7
UNDER DEV

Deepsurgery Medical Instrument
Dessight Biomedical
F.MED / Ｆ．ＭＥＤ株式会社
Gibbon Medical
KouTech / 昂泰微精
Qoly Corporation： クオリィ株式会社
Sony

1
ACQUIRED

True Digital Surgery

1
UNCERTAIN

Elytra Technologies

Ophthalmic

Representative examples



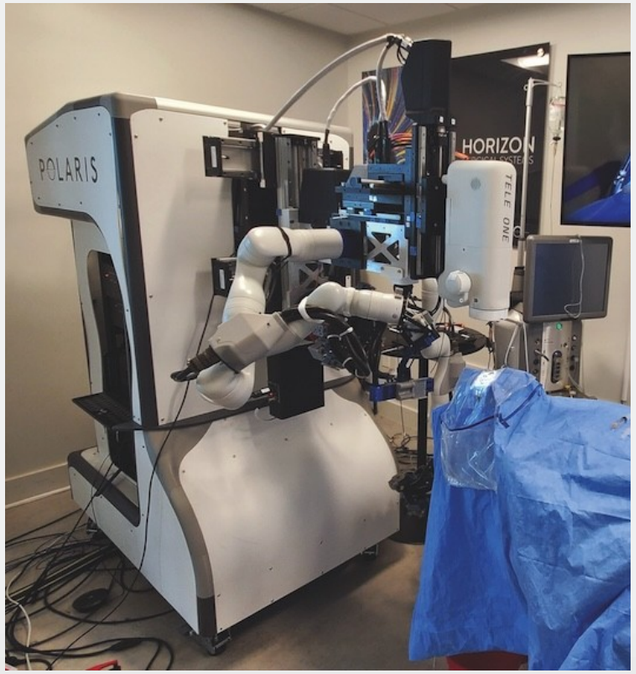
PRECEYES



Acusurgical



FORSIGHT ROBOTICS



HORIZON SURGICAL SYSTEMS

Ophthalmic

27 robots from 27 companies

5
ON MARKET

Belkin Vision
Beyeonics
LENSAR
PRECEYES Medical Robotics
Riverfield Inc.

20
UNDER DEV

Acusurgical
Alcon
Dessight Biomedical
ForSight Robotics
Horizon Surgical Systems, Inc.
Impel IP LLC
J&J Surgical Vision
Keranova
Mingche Biotechnology / 明澈生物科技
Mynutia
Ocutrx Vision Technologies
Ophthorobotics AG
Optorobotics S.r.l.
RIMSCIENCE Co., Ltd.
Robotrak
Suroviz
TIROMU / 瞳沐医疗
VisioSurgical Solutions
Weimou Medical / 微眸医疗
Xianwei Medical Technology

2
DEAD

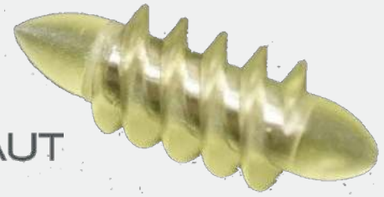
Medical Robotic Technologies B.V.
NSK Ltd. / 日本精工

Micro / Nano Bots

Representative examples



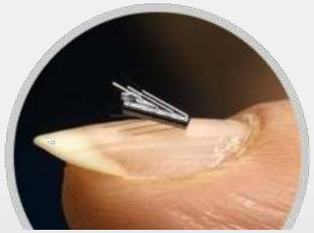
ARTEDRONE

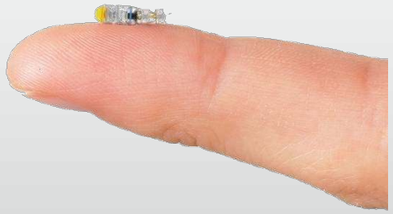


BIONAUT LABS

microbot medical

ViRob
Life in motion





ROBEAUTÉ

Micro / Nano

12 robots from 11 companies

10
UNDER DEV

Bionaut Labs
IMSystem / 아임시스템
InterMag
Interventional Systems
Koswire / 코스와이어
Macheon AI
MagnebotiX
Robeauté
Symphony Robotics
ValueBioTech S.R.L.

1
ACQUIRED

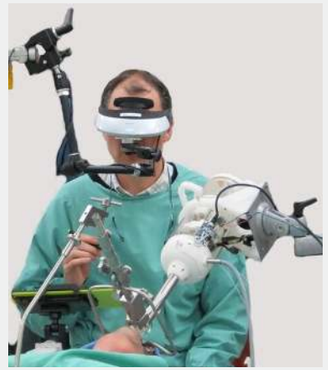
ArteDrone

Otolaryngology

Representative examples



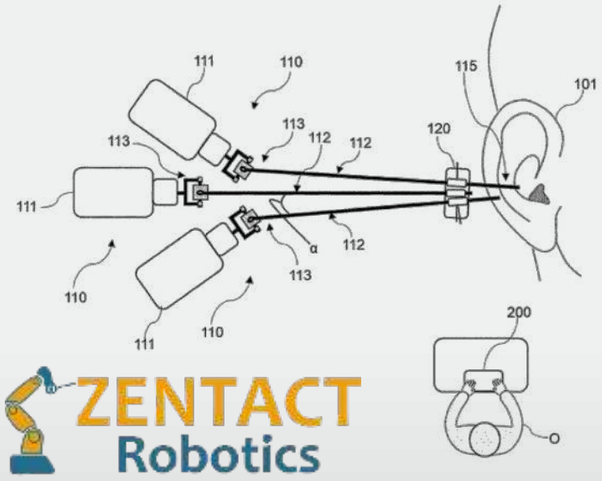
iota



Monarch TECHNOLOGIES



Medrobotics



ZENTACT Robotics



思哲睿医疗 SAGEBOT MEDICAL

Otolaryngology

18 robots from 17 companies

4
ON MARKET

Collin Medical
Galen Robotics
iotaMotion
MED-EL / CAScination

10
UNDER DEV

Amarob Technologies
Borns Medical Robotics / 博恩思
Cochlear Limited
NavLab Inc. / NavLab Holdings II
Sagebot / 思哲睿
Tamar Robotics Ltd.
Taurean Surgical, Inc.
Tianchen Medical
Weigao Group / 威高集团
Zentact Robotics

3
DEAD

Brain Navi Biotechnology / 鈦隼生物科技
InnovoTone LLC
URS

Exits, failures, and the money

Who was acquired · who died · who buys · exit economics · SPACs · what changed in 2026 · where the money sits

Robots or companies whose status changed: acquired

43 companies (45 robot rows) carry Acquired status, Computer Motion (2003) to Robocath (2026)

A – C

ArteDrone · Asensus Surgical · Auris Health · Blue Belt Technologies · Caranx Medical · Cardan Robotics · Computer Motion · Corindus · Covidien · CyberHeart

D – K

Denso Corporation · Endocontrol SA · Excelsius Surgical · Fusion Robotics · Gomtec · Great Belief International Limited · Human Xtensions · Innomedic · JNJ Ethicon Endo-Surgery RAD program · KAT Robotics · KB Medical

L – O

LinkX Robotics · Mako Surgical Corp. · Mazor Robotics · MDA · Medical Surgery Technologies · Medineering · Mobius Imaging · Monogram Technologies · OMNIlife Science · OrthoSpin Ltd. · Orthotaxy SAS · Ostesys

R – V

Relievant Medsystems · RoboCath · Ruby Robotics · Sensus Healthcare, Inc. · SOFAR S.p.A. · TransEnterix · True Digital Surgery · Tuge Healthcare · Verb Surgical · VytronUS, Inc.

Robots or companies whose status changed: dead

64 dead companies/programs (66 robot rows), up from 48 in January, several parents remain active elsewhere

A – C

Acrobot · Advanced Osteotomy Tools · Aeon Scientific · Agile EndoSurgery · American GNC Corporation · Avatera Medical · AVRA Medical Robotics · Bio-Medical Engineering · Borns Medical Robotics · Boston Scientific · Brachium · Brain Navi Biotechnology · Catheter Precision, Inc. · Centauri Robotic Surgical Systems · Comau SpA · CranioMedix Inc. · Cyber-Implants LLC

E – L

Euphrates Vascular · FocalTherics · GuidaBot LLC · Hansen Medical · Human Robotics S.A. de C.V. · iin medical · InnovoTone LLC · Intuitive Surgical · Invendo Medical · Invivo Robotics · ISIS · Lapara Surgical · LG Electronics · Longhui Medical

M – R

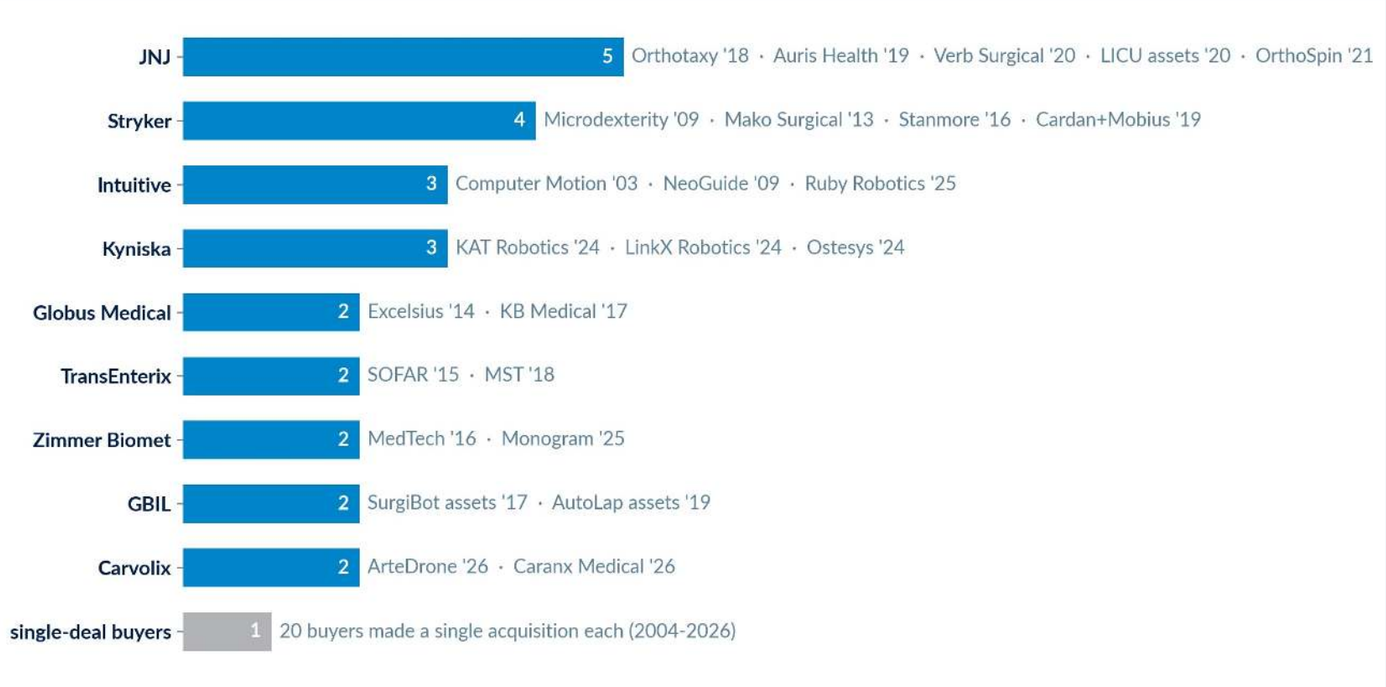
MDA · MDA Space Ltd. · Medical Robotic Technologies B.V. · Medrobotics Corporation · MichelAngelo RobotScope · Moray Medical · Motus GI Medical Technologies · National Semiconductor · NeoGuide Systems · NeuroArm · NSK Ltd. · Omniboros · ortoMaquet GmbH · Panasonic Holdings · Polymer Robotics Inc. · Revision Robotics Surgical Innovations, LLC · Robopsy · Roboscopy Medical Ltd. · Rubicon Spine, LLC

S – Y

Sagitta Robotics Limited · Samsung Electronics · Seiko Epson · Siemens Healthineers Endovascular Robotics · Signature Robot Ltd · SPI Surgical, Inc. · St. Jude Robotic Catheter · Starpax Biopharma · Synaptive Medical · Telehealthrobotics · URS · Vicarious Surgical Inc. · XACT Robotics Ltd. · Yikangtai Medical Technology

Exits concentrate in a few repeat buyers

Acquisitions per buyer with targets and deal years; single-deal buyers aggregated. Sources: deal press releases and SEC filings; collected 8 Jul 2026



45 RESEARCHED DEALS

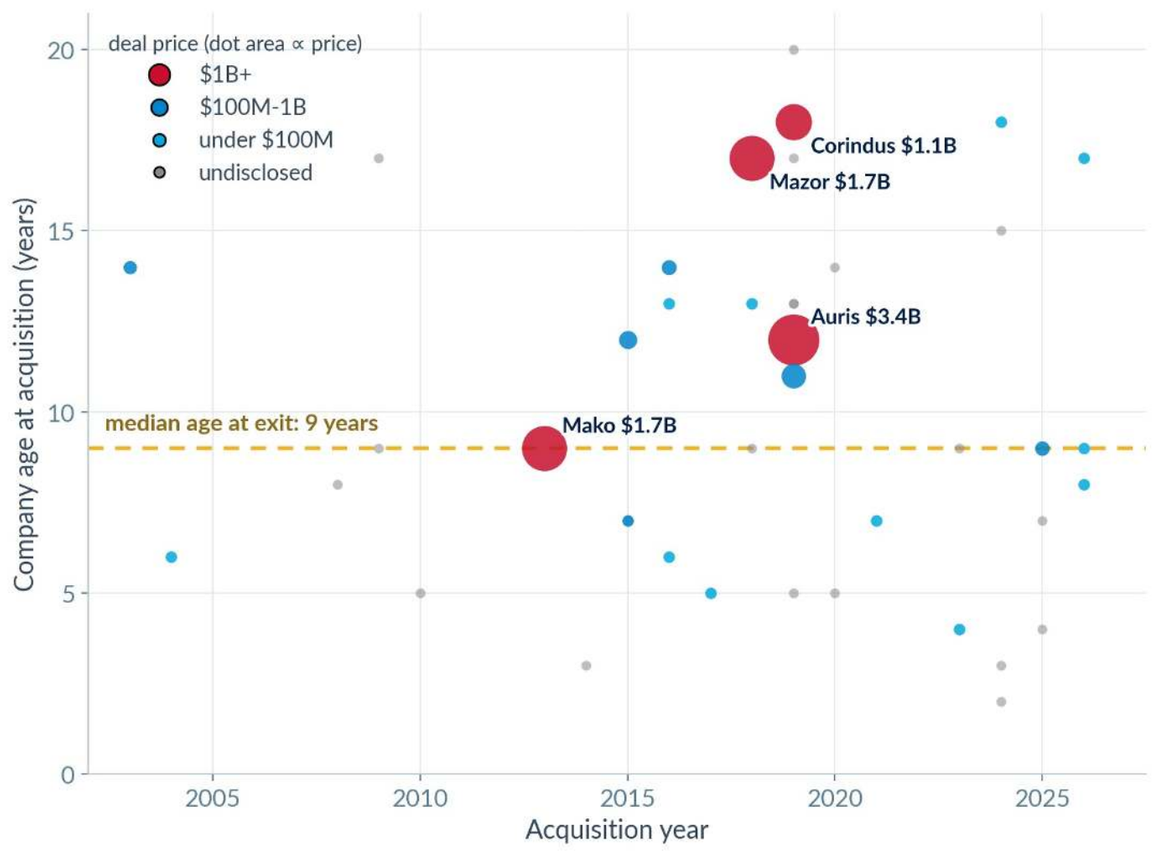
9 repeat buyers account for 25 of 45 acquisitions

Busiest: JNJ, Stryker, Intuitive, Kyniska

Repeat buying concentrates in orthopedics/spine and soft tissue

Acquisition typically takes about a decade

Age at acquisition vs deal year. Sources: press releases and SEC filings; current as of 8 Jul 2026



MEDIAN AGE AT EXIT: 9 YEARS

45 acquisitions researched, 2003 to 2026; 41 with a known company age are plotted

Disclosed prices span $5M to $3.4B; billion-dollar deals labeled

Gray dots: undisclosed price. Asset deals and mergers carry * in the table

Exit economics: the full record (1 of 2)

Raise-to-exit economics, UD = undisclosed, * = special aspects; maintained in the deck data file; current as of 8 Jul 2026

COMPANIES	FOCUS AREA	COUNTRY	INVESTMENT($M)	# ROUNDS	ACQ PRICE ($M)	ACQ YEAR	AGE	# INVESTORS	ACQUIRER
ACROBOT	orthopedic	UK	5	1	UD	2010	~5	3	Stanmore
ARTEDRONE*	neurovascular	France	~23	UD	~12.5	2026	9	1	Carvolix
ASENSUS	abdominal/thoracic	US	484	18	95	2024	18	12	Karl Storz
AURIS HEALTH*	abdominal/thoracic	US	733	8	3400	2019	12	13	JNJ
AVATERA MEDICAL*	abdominal/thoracic	Germany	~270	UD	UD	2023	9	1	Tennor Internationa…
BLUE BELT TECHNOLOGIES*	orthopedic	US	45	2	275	2015	12	4	Smith+Nephew
CARANX MEDICAL*	cardiovascular	France	~25 + 5 grants	UD	~18	2026	8	2+	Carvolix
CARDAN ROBOTICS + MOBIUS…	spine	US	45 + ~82	2	500	2019	11	1	Stryker
COMPUTER MOTION*	abdominal/thoracic	US	37(IPO) + 22+ post-IPO	UD	~150	2003	14	UD	Intuitive
CORINDUS	endovascular	US	134	7	1100	2019	18	4	Siemens Healthineers
CYBERHEART	cardiac	US	24	4	UD	2019	13	3	Varian
DENSO / IARMS*	surgeon arm-support…	Japan	n/a	n/a	UD	2018	n/a	n/a	Toho Technology
ENDOCONTROL	abdominal/thoracic	France	6	1	UD	2019	13	2	Canady Robotics
ENDOVIA MEDICAL	abdominal/thoracic	US	12	4	5	2004	6	~3	Hansen Medical
EXCELSIUS SURGICAL*	spine	US	5.5	1	UD	2014	3	UD	Globus Medical
FUSION ROBOTICS…	spine	US	UD	UD	55	2023	4	UD	ATEC
GOMTEC*	abdominal/thoracic	Germany	UD	UD	UD	2015	7	UD	ABB
HANSEN MEDICAL	endovascular	US	116	5	240(IPO) + 80	2016	13	5	Auris
INNOMEDIC*	orthopedic	Germany	UD	UD	UD	2008	~8	UD	Synthes
KAT ROBOTICS*	orthopedic	France	~0.5	1	UD	2024	2	1	Kyniska Robotics
KB MEDICAL*	spine	Switzerland	7	4	~31.5 + 4.9 contingent	2017	5	8	Globus Medical
LINKX ROBOTICS*	orthopedic	France	~0.5	1	UD	2024	3	1	Kyniska Robotics
MAKO SURGICAL*	orthopedic	US	95	5	51(IPO) + 1700	2013	9	7	Stryker

WHAT THE RECORD SHOWS

24 of 45 deals disclose a price; median $79M

4 deals of $1B+ sit atop a long tail: most exits are under $100M

21 deals without a disclosed price cluster in the early years and small tuck-ins

7 asset-level deals (SurgiBot, AutoLap, REMI, Sculptura) show IP outliving its company

Spine and orthopedics account for the largest share of exits

UD = undisclosed or unknown · n/a = not applicable (asset deal or corporate program) · ~ = approximate · * = special aspects (see data file) · IPO = initial public offering · CVR = contingent value right

Exit economics: the full record (2 of 2)

Raise-to-exit economics, UD = undisclosed, * = special aspects; maintained in the deck data file; current as of 8 Jul 2026

COMPANIES	FOCUS AREA	COUNTRY	INVESTMENT($M)	# ROUNDS	ACQ PRICE ($M)	ACQ YEAR	AGE	# INVESTORS	ACQUIRER
MAZOR SURGICAL TECH	spine	Israel	70	3	1700	2018	17	6	Medtronic
MEDICAL SURGERY TECH*	abdominal/thoracic	Israel	14	2	~31	2018	13	6	TransEnterix
MEDINEERING*	neuro	Germany	UD	2	UD	2019	5	4	Brainlab
MEDTECH	neuro & spine	France	2	1	188	2016	14	1	Zimmer Biomet
MICRODEXTERITY*	orthopedic	US	UD	UD	UD	2009	17	UD	Stryker
MONOGRAM TECHNOLOGIES*	orthopedic	US	~75 + 30(IPO) + post-IPO	10	177 + CVR	2025	9	crowd…	Zimmer Biomet
NEOGUIDE	gastroenterology	US	39	2	UD	2009	9	9	Intuitive
OMNILIFE SCIENCE	orthopedic	US	32	3	UD	2019	20	2	Corin Group
ORTHOSPIN*	orthopedic	Israel	8	~3	79.5	2021	7	2	DePuy Synthes (JNJ)
ORTHOTAXY*	orthopedic	France	UD	UD	UD	2018	9	UD	JNJ
OSTESYS*	orthopedic	France	UD	UD	UD	2024	15	2	Kyniska Robotics
RESTORATION ROBOTICS*	hair	US	129	9	25(IPO) + stock	2019	17	5	Venus Concepts
ROBOCATH*	vascular intervention	France	~88	6	20 + 25 earnout	2026	17	10+	Stereotaxis
RUBY ROBOTICS*	autonomous point-of-c…	US	5.6	~2	UD	2025	4	4	Intuitive
SENSUS HEALTHCARE…	robotic IORT…	US	n/a	n/a	15	2022	n/a	n/a	Empyrean Medical…
SOFAR*	abdominal/thoracic	Italy	UD	UD	100	2015	~7	1	TransEnterix
STANMORE IMPLANTS*	orthopedic	UK	34	UD	52	2016	6	UD	Stryker
TRANSENTERIX / AUTOLAP…	abdominal/thoracic	US	n/a	n/a	47	2019	n/a	n/a	GBIL
TRANSENTERIX / SURGIBOT…	abdominal/thoracic	US	n/a	n/a	29	2017	n/a	n/a	GBIL
TRUE DIGITAL SURGERY*	digital surgical…	US	~49	UD	UD	2025	7	5	B. Braun
VERB SURGICAL*	abdominal/thoracic	US	~500	1	UD	2020	5	2	JNJ
VYTRONUS / LICU ASSETS*	cardiac EP	US	~115	7	UD	2020	14	5	JNJ

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Special Purpose Acquisition Corporations (SPACs)

The SPAC cohort, and how it went: Memic SPAC failed, re-emerged as Momentis; Vicarious delisted Mar 2026, stockholders approved dissolution Jul 2026



Delisted Mar 2026 · stockholders approved dissolution Jul 2026

debut on Wall Street on Monday. The company closed its previously announced business combination with D8 Holdings Corp. last week, gathering $220 million in gross proceeds. The deal, announced this past April, valued the combined company at an enterprise value of approximately $1.1 billion.

memic

breaking surgical boundaries



MOMENTIS
SURGICAL

Israel medical device company **Memic Innovative Surgery Ltd.** today announced the signing of a definitive agreement for a SPAC merger with MedTech Acquisition Corporation (Nasdaq: MTAC). The merged company will have $360 million cash and an equity value of more than $1 billion.

Robots or companies whose status changed in 2026

Exits validate the category; the flameouts show that approval ≠ commercialization

EXITS & WINS

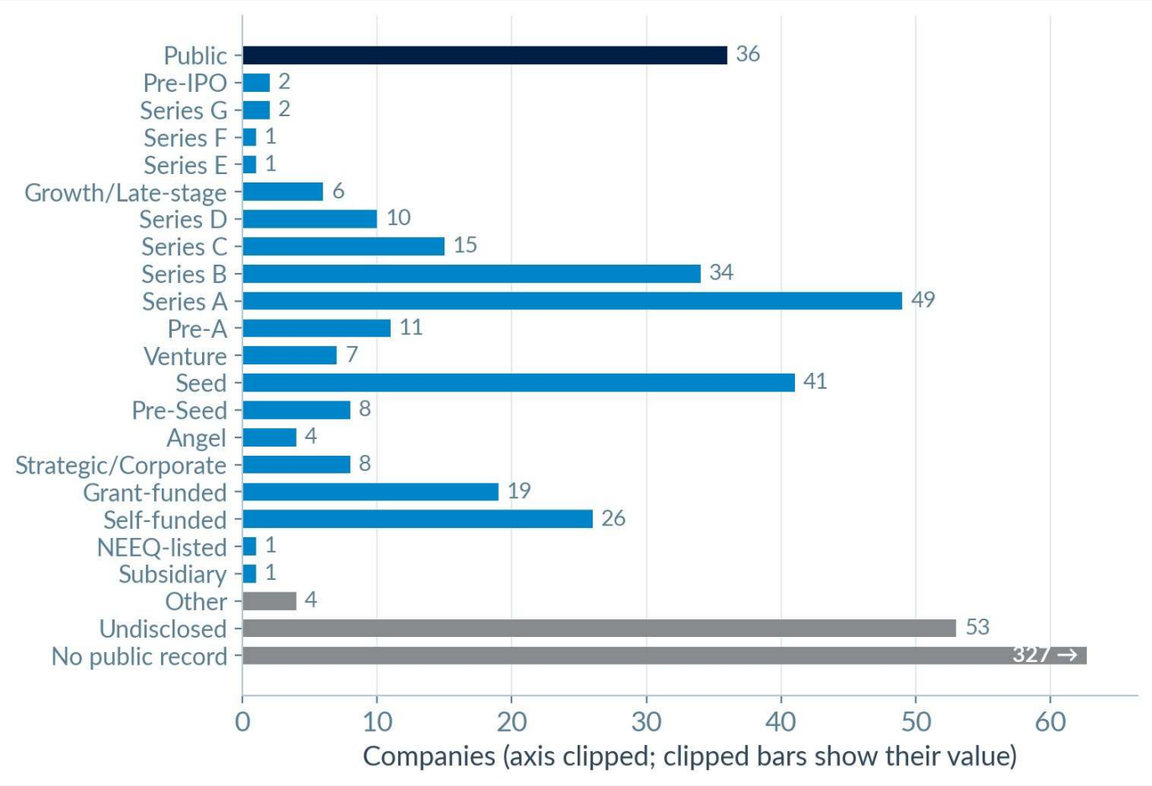
- **TrueHealth / 真健康 → HKEX IPO (Jun 30)**
  +217% day one, ~HK$14.3B cap, ~1,800× oversubscribed, capital is concentrating on the category leader
- **Edge Medical / 精锋 → Hang Seng Composite**
  Index inclusion Jun 8; FY2025 revenue RMB 456M (+185%); overseas installs now exceed domestic
- **Stereotaxis × Robocath**
  Endovascular consolidation (Apr 2026)
- **Ruichu Robotics → HKEX A1 filed (May 27)**
  Pre-revenue lung-puncture robot chasing the TrueHealth window

WOUND DOWN / DELISTED

- **Vicarious Surgical (US)**
  NYSE delisting Mar 3 (cap <$15M) → OTC as RBOT; first 2026 Western flameout; stockholders approved dissolution Jul 2026 (liquidation)
- **Asensus / Senhance + Luna (US/DE)**
  Karl Storz discontinues both robots and retires the brand, a ~20-year program ends (strategic wind-down, not bankruptcy)
- **Longhui Medical / 龙慧 (China)**
  NMPA approval Sep 2024 → bankruptcy Jan 2026: approval ≠ commercialization
- **Microya / 微亚 (China)**
  Entire IP + assets auctioned from ~RMB 1.29M starting bid

49% of the field has no public funding story

Funding stage per company (normalized): 666 of 677 companies, dead ones included; the 11 whose stage reads Acquired are in the exits section



327 NO PUBLIC FUNDING RECORD

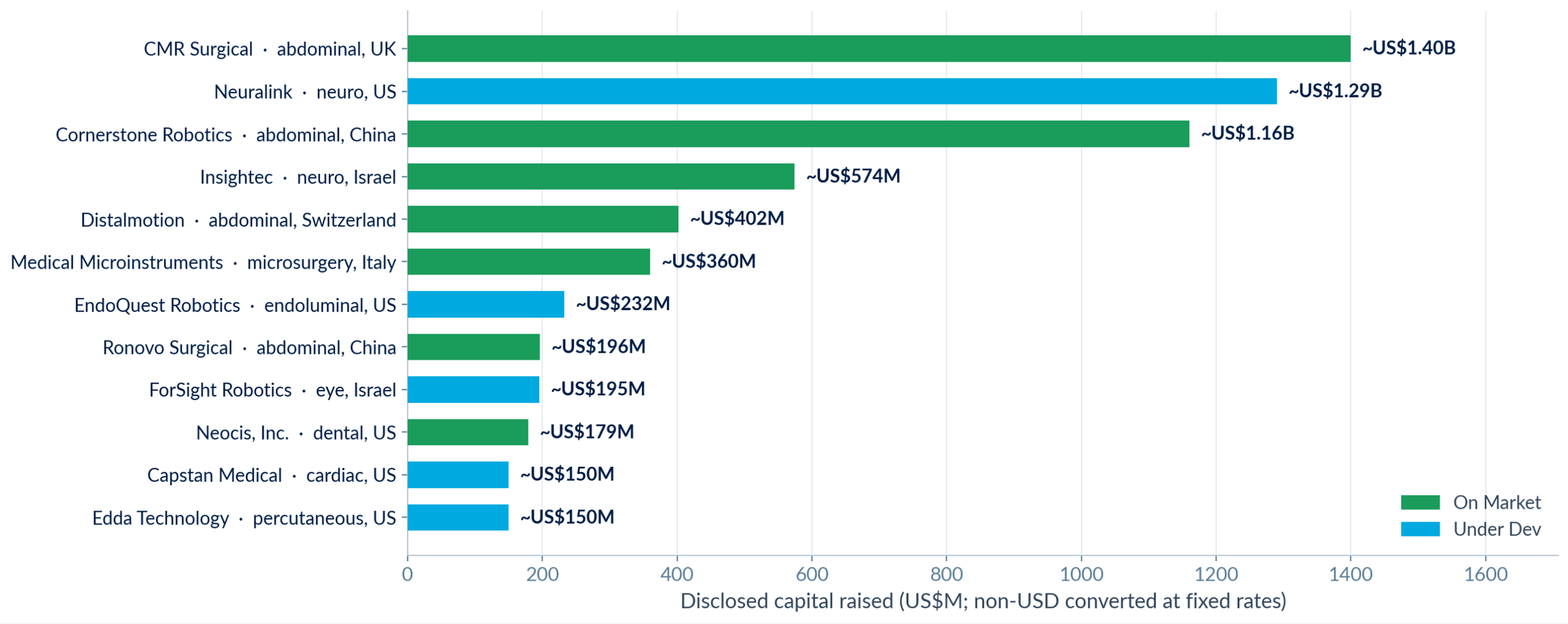
49% of companies; mostly Chinese patent-only finds, institutes, and quiet SMEs

36 public companies bookend the other extreme

164 companies disclose a raise amount; Seed-to-B dominates the private tail

Best-funded private players

Ranked by disclosed capital raised (July 2026), per the master roster



Not shown: self-funded strategics & public leaders, Intuitive, Medtronic (Hugo), J&J (Ottava/Monarch), Stryker (Mako), MicroPort MedBot, Edge Medical (IPO $154M Jan 2026), SS Innovations, PROCEPT. Also MMI ($110M+ Series C) and Edda Technology ($150M strategic at ~$750M valuation).

Examples of insights from assessing a range of commercial efforts

What the collection teaches: form factors, topologies, consoles, controls

Abdominal/thoracic form factors

Representative examples, single-port remains the exception











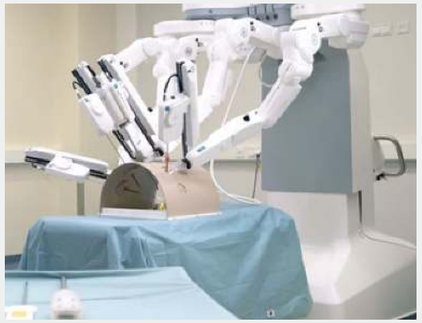
















15 Single port

116 Multi port

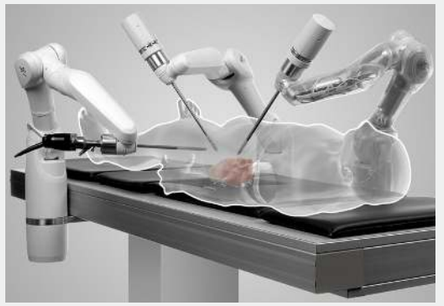
Varying topologies

Representative examples

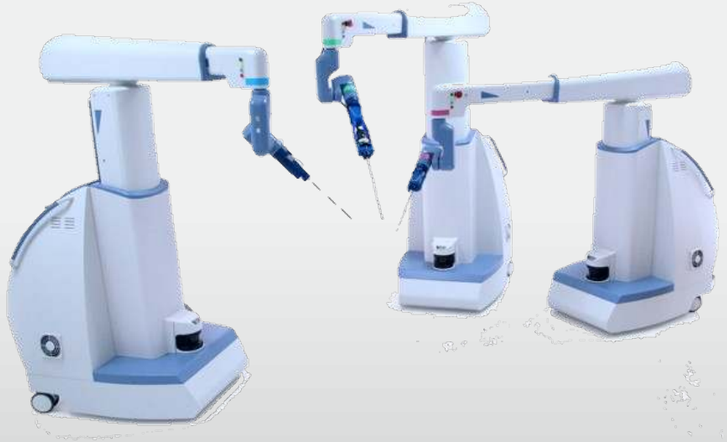












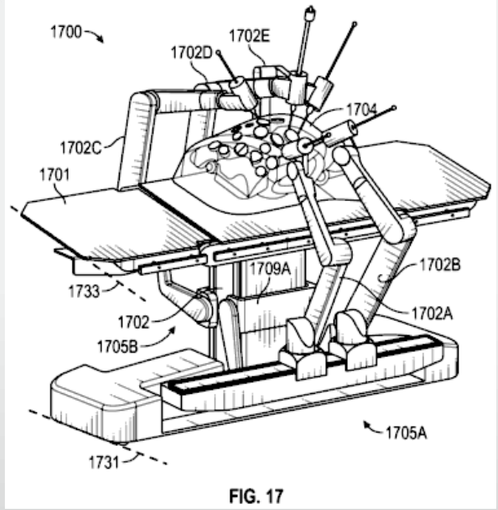


FIG. 17

Varying consoles

Representative examples



INTUITIVE



CMR SURGICAL



revo™ surgical solution



TITAN MEDICAL





ASENSUS SURGICAL



Medtronic



Medicaroid



Distalmotion

MicroPort

SS INNOVATIONS





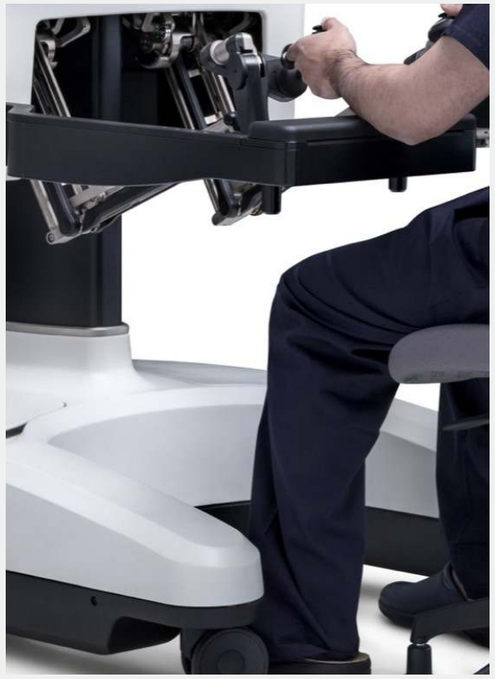
avatera

Varying hand and foot controls

Representative examples



INTUITIVE



CMR SURGICAL



Medtronic



TITAN MEDICAL

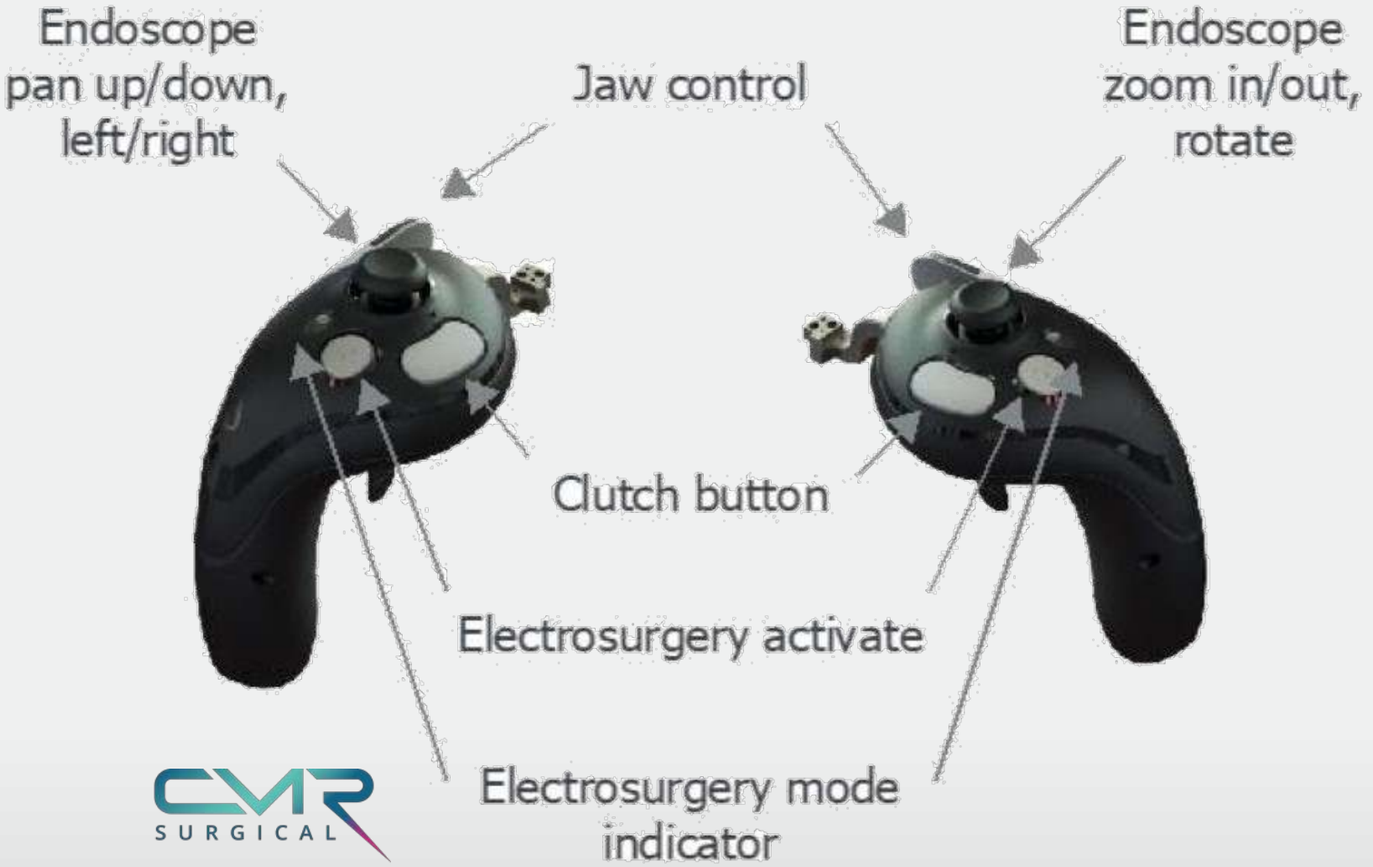


Distalmotion

Simple squeeze vs complex handle

Those missing controls from the squeeze handle live somewhere...





Novel approaches

Representative examples

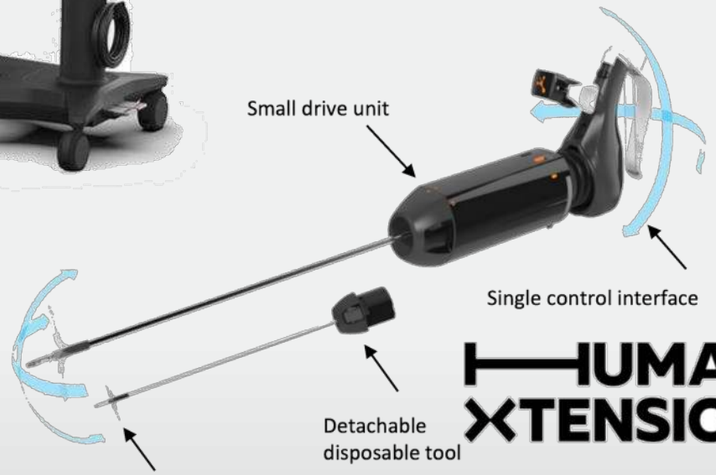
VICARIOUS SURGICAL INC











HUMAN XTENSIONS™

MOMENTIS SURGICAL™





VIRTUALINCISION

Major trends

Where the field is stuck, and where it should go next

- **Still too many robots in the same specialties**
  …with similar approaches, and still too many full-platform solutions
- **Control-interface variety is a problem**
  Physical and visual control interfaces differ widely from platform to platform
- **Just roboticizing what is done by hand is not enough**
  First autonomous execution of a procedure phase supports this (Science Robotics, 2025)

GOING FORWARD

Opportunity for existing platform enhancement

- Operate in smaller and/or lower-resourced facilities
- Pay-per-use rather than capital + tool/drape purchases
- Lower capital cost · smaller footprints
- Wear-based rather than number-of-uses-based tool replacement
- Higher-power-density small actuators will change device configurations
- Judicious use of soft actuators as well as live hinges
- Legal frameworks and monetization of telesurgery

Machine learning

Everywhere, expected, penalized if absent; marketing still runs ahead of reality

- **The barrier to entry keeps dropping**
  Technical and data-availability thresholds continue to fall; foundation models increasingly sit beneath specialist data
- **Most applications will be under the hood**
  Optimizing the robot itself; more small, inexpensive sensors coupled with ML/AI video
- **Simulation for training content**
  There is a strong case for simulation in creating training data and content

LEVELING OUTCOMES ACROSS LOCATION & SURGEON

Enhanced surgical-site situational awareness

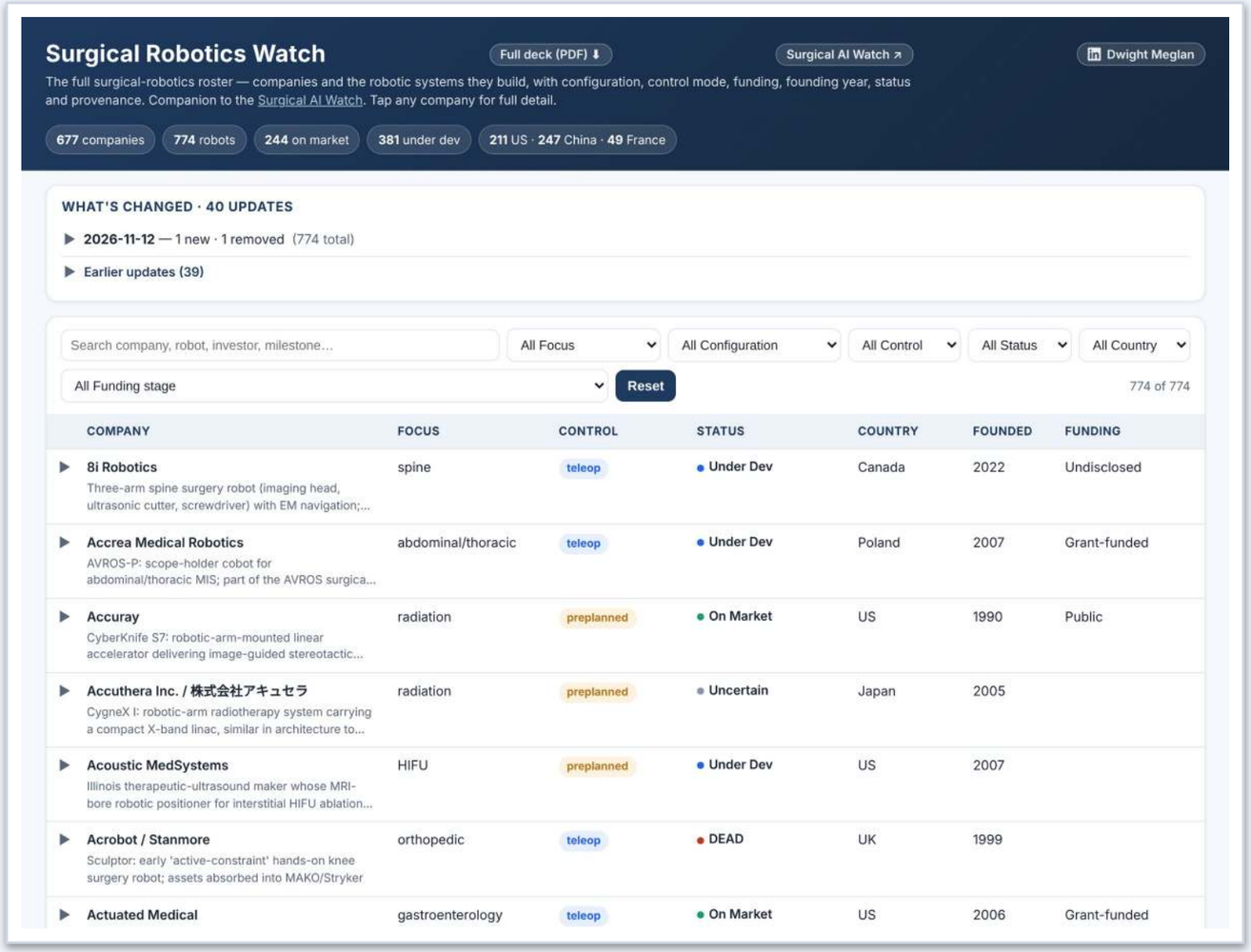
- Contextualization relative to best practices
- Ready availability of ad-hoc mentorship
- Subtask enhancement / automation
- Removing the need for technicians to run things

- Autonomous cholecystectomy phase demonstrated ex vivo, 8/8 (Kim et al., Science Robotics 2025)

Tools like Claude Code are accelerating and broadening use cases, but will need to be thorough in verification: they very quickly make things that look great but have unclear provenance

Surgical Robotics Watch, a live companion dashboard

An updating web version of the data underlying this deck, search, filter, sort, and drill into any company



LIVE AT

dmeglan.github.io/surgical-robot-companies-dashboard

WHAT IT IS

A self-contained, mobile-responsive dashboard generated from the same master spreadsheet behind this deck

WHAT YOU CAN DO

- Full-text search across company, robot, investor, milestone
- Filter by focus area, configuration, control, status, country, funding stage
- Sortable table; tap any company for full detail with milestones and provenance
- STAYS CURRENT

  Rebuilt with every roster refresh (twice-weekly watch); a "what changed" panel accompanies every update



Questions / Discussion?